

Target-Fixed Physical Role Necessity for the Realized Neutrino Oscillation Interface

Interaction–Propagation Nonalignment, Boundary-Visible Response Multiplicity, and Residual Role Closure

Scope of the result

The analysis fixes the experimentally realized source–propagation–detection interface while leaving its physical completion unspecified. The theorem identifies only the physical work that any realization of that interface must perform. No neutrino mass mechanism, PMNS matrix, Dirac/Majorana choice, sterile sector, generalized-symmetry completion, new representation, or candidate particle is assumed in the necessity argument.

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Abstract

Neutrino flavor transformation is conventionally represented by a nontrivial mixing matrix and nondegenerate propagation phases. This paper asks a logically prior question: what physical work is required by the realized source–propagation–detection interface before any microscopic carrier or parameterization is specified? The fixed target combines the known electroweak charged-current boundary architecture with primary experimental evidence for nontrivial flavor transfer, baseline/energy-dependent modulation, and two spectrally resolved reactor-response scales in the same operational electron-antineutrino boundary channel. The completion responsible for those observations is left open.

A carrier-neutral propagation interface is formulated at channel level using a record-relevant state space $\mathcal{H}_\nu$, charged-current boundary faces generated by P_α , source states ρ_α , a lawful family of completely positive trace-preserving maps $\mathcal{E}_\lambda$, and transition kernels

$$K_{\beta\alpha}(\lambda) = \text{Tr}(P_\beta \mathcal{E}_\lambda(\rho_\alpha)) .$$

The first theorem shows that certified cross-boundary transfer requires interaction–propagation nonalignment: at least one charged-current boundary face must fail to remain invariant under the complete propagation channel. The familiar commutator condition $[P_\alpha, U_\lambda] \neq 0$ is recovered as the unitary representative of this channel-native requirement.

The second theorem isolates a distinct response burden on the common reactor $\bar{\nu}_e$ survival interface. Each experiment contributes a finite phenomenological phase-frequency interval together with its published response model and nuisance treatment. After quotienting only by source-authorized calibration, environment, and analysis-coordinate transformations, KamLAND and Daya Bay determine two inequivalent visible response-mode classes. Writing $\mathfrak{M}_{\text{vis}}$ for the resulting mode-class set and $\rho_{\text{vis}} := |\mathfrak{M}_{\text{vis}}|$ for its finite multiplicity, the target requires $\rho_{\text{vis}} \geq 2$. A one-mode nonaligned propagator provides an explicit counterexample to the converse implication, so visible response multiplicity is strictly stronger than nonalignment rather than identical to it.

Both coordinates are invariant under simultaneous lawful redescription and cannot be generated by a weak-basis choice, rephasing, eigenvector labeling, or PMNS chart. The minimal renormalizable Standard Model with massless neutrinos supplies the source/detection boundary and the general physical role architecture, but it supplies neither boundary-sector transfer nor a nontrivial visible response-mode class. The resulting target-specific residual role complex is

$$R_{\text{res}}(C_\nu, T_\nu^{\text{osc}}; D_{\text{SM}}^{\nu 0}) = (\{r_{\bowtie}, r_\Delta\}, r_\Delta \succ r_{\bowtie}) \neq \emptyset,$$

where r_Δ denotes the boundary-visible response-multiplicity obligation $\rho_{\text{vis}} \geq 2$. The ordering records dependence between pieces of physical work, not carrier cardinality. The result is therefore an unoccupied physical-role obligation relative to the minimal massless-neutrino Standard Model, not a prediction of a particle, mass mechanism, mixing matrix, ordering, or absolute scale.

Main result

Conditional on the explicitly stated phenomenological response assumptions and the fixed source–propagation–detection comparison class, the realized neutrino oscillation target requires a quotient-stable interaction–propagation nonalignment role and a strictly stronger boundary-visible response-multiplicity role with $\rho_{\text{vis}} \geq 2$; the minimal massless-neutrino Standard Model occupies neither target-specific coordinate.

Scope of the claims

Status	Content	Scope
Established here	Channel-native no-transfer/nonalignment; lawful-redescription invariance; finite visible-response multiplicity under the stated empirical response model; same-target deletion; strict role non-equivalence; minimal-target nonoccupancy; residual-role composition.	Relative to the target and comparison assumptions stated explicitly below.
Imported theorem	AASC kernel and anti-weakening; realized-physical-interior roles; Role-Occupancy Closure discipline; simultaneous-constraint warning; Standard Model and symmetry target separation.	No imported theorem supplies the neutrino-specific residual role conclusion or a carrier identity.
Empirical input	Operational flavor tagging, nontrivial flavor transformation, path/energy modulation, and at least two separated reactor response-frequency classes.	Primary experimental analyses instantiate the target; their microscopic interpretation is not used to choose a completion.
Conditional input	Effective channel description and the collaboration-level phenomenological response models used to extract finite phase-frequency intervals.	A common-boundary reanalysis that invalidated this response-model class or collapsed the two intervals to one lawful class would reopen r_{Δ} .
Open	Physical occupant, field content, mass character, state-space dimension, global/topological completion, numerical parameters, ordering, absolute scale, and further observable predictions.	Not claimed in this paper.

Table 1: Scope and status of the principal claims.

Nonclaims

The paper does not derive neutrino masses; does not infer that the occupant is a particle; does not select Dirac, Majorana, pseudo-Dirac, sterile, radiative, topological, defect, or nonunitary completion; does not derive a PMNS matrix or its numerical entries; does not predict mass ordering, the absolute neutrino mass scale, or an experimental rate; and does not interpret $\rho_{\text{vis}} \geq 2$ as exactly two masses, gaps, fields, couplings, microscopic eigenmodes, or model-independent Fourier atoms.

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1 Introduction and theorem strategy

1.1 The role-necessity problem

The analysis begins before neutrino-mass model building. Its target is neither a Lagrangian term nor a set of new fields nor a fitted parameter matrix. The target is the experimentally realized source–propagation–detection interface described in section 2. The question is which target-specific pieces of physical work cannot be removed while that interface is held fixed.

This ordering is essential. Role-Occupancy Closure requires the target role set to be fixed before the admissible residue is inspected; otherwise the number and type of roles can be back-fitted to whichever candidates happen to survive [16, Def. 3.5, Thm. 6.1, Cor. 6.2]. The generation-cardinality result provides the same methodological precedent: the required role load is fixed before the open cardinality is closed [14]. Here the same discipline is applied to an open physical-completion problem.

Definition 1.1 (Target-forced physical role). *Let C be fixed target data and T the physical target they specify. A target-forced physical role is a target-relative work item r satisfying all of the following:*

- (i) *the work is required for realization of T under C ;*
- (ii) *deleting the work while preserving the other target anchors destroys at least one declared target record or makes the target ill-typed;*
- (iii) *the work and its deletion consequence are invariant under the lawful equivalence fixed by C ;*
- (iv) *the definition of the work does not name, presuppose, or privilege an eventual occupant.*

Definition 1.2 (Known occupancy and residual role complex). *Let $R_T(C, T)$ be the target-forced role set and let $R_{\text{occupied}}^{\text{known}}(C, T)$ be the subset for which the known comparison substrate supplies an explicit same-target occupant. The residual role content is*

$$R_{\text{res}}(C, T) := R_T(C, T) \setminus R_{\text{occupied}}^{\text{known}}(C, T).$$

An unknown occupancy status is not a negative result and therefore cannot be placed in the residual set merely by absence of evidence.

These definitions separate three questions that are often conflated: what physical work is required, what known structure already performs it, and what completion ultimately realizes any unoccupied work. The present paper answers only the first two.

1.2 Proof strategy

The proof combines fixed-target deletion with exhaustion of same-target redescrptions. The AASC kernel supplies the four governance roles required for non-degenerate identity-preserving construction—admissibility, standing, reference, and irreversibility—and its fixed-domain exhaustion theorem blocks same-domain weakening by selector, repair, or silent target drift [15, Thm. 2.34, Thm. 6.20]. The AMetric boundary blocks a numerical score, likelihood, canonical representative, or discrete index from authorizing boundary standing [18, Thm. 5.26, Thm. 14.2]. These results do not establish any neutrino-specific conclusion; they constrain the admissible form of the necessity argument.

Theorem 1.3 (Carrier-neutrality of role necessity). *A target-role definition that contains a candidate-specific carrier coordinate—including a neutrino mass term, a PMNS entry, a right-handed-neutrino field, a Weinberg coefficient, a sterile-state label, a defect category, or a preferred numerical parameter—does not establish target-forced role necessity. It establishes at most necessity relative to a preselected completion class.*

Proof. A candidate-specific coordinate is not fixed by the source–propagation–detection target itself. Introducing it before the role set is fixed either changes the target data or inserts a selector into the role definition. In the first case the conclusion concerns a narrower completion target; in the second it is circular because the intended occupant has been built into the work it is said to occupy. Under the fixed-domain and AMetric assumptions, neither move yields a same-target derivation of an occupant-neutral role. $\square$

Corollary 1.4 (No goal-directed role design). *The prospect of obtaining a novel experimentally testable prediction cannot function as a premise in the role-necessity argument. A role survives only if the fixed target loses standing when that role is deleted.*

1.3 Logical order of the analysis

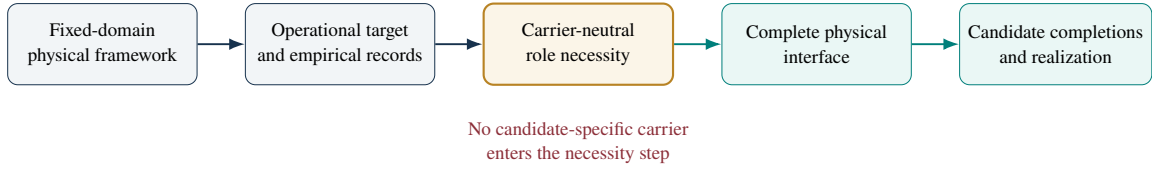


Figure 1: Logical order of the analysis. The target and empirical records are fixed before the carrier-neutral role analysis; candidate generation and realization questions remain subsequent problems.

The argument proceeds in five stages. First, the empirical and operational target is fixed. Second, the source–propagation–detection interface is expressed without assuming a microscopic completion. Third, target-forced roles are identified through invariant deletion tests. Fourth, the known minimal Standard Model comparison fiber is tested for occupancy only after those roles have been defined. Finally, the residual role complex is formed as a set difference. No candidate completion is selected at any stage of this chain.

1.4 Theorem boundary

The strongest result established below is

$$R_{\text{res}}(C_{\nu}, T_{\nu}^{\text{osc}}; D_{\text{SM}}^{\nu 0}) = (\{r_{\triangleright\Delta}, r_{\Delta}\}, r_{\Delta} \succ r_{\triangleright\Delta}),$$

where $r_{\triangleright\Delta}$ is channel-native interaction–propagation nonalignment and r_{Δ} is the boundary-visible response-multiplicity requirement $\rho_{\text{vis}} \geq 2$. The order symbol records a dependence between pieces of physical work; it does not imply two particles, two fundamental parameters, or two distinct carrier objects.

The result is a necessity statement, not a sufficiency theorem. It leaves open whether an admissible completion exists, whether one or many carrier classes survive, whether the eventual realization is local matter or another physical structure, whether role occupancy is unique, and whether any further observable consequence is experimentally accessible.

2 Fixed target and empirical evidence

2.1 The target tuple

The target is the realized low-energy neutrino source–propagation–detection interface, not the standard mass–mixing representation of that interface. It is specified by

$$T_{\nu}^{\text{osc}} = \left(D_{\text{EW}/\ell}^{\text{fixed}}, \mathcal{B}_{\text{CC}}, \mathcal{P}_{\text{rec}}, \mathcal{E}_{\text{osc}}, \mathcal{Q}_{\text{phys}}, C_{\text{open}} \right). \quad (1)$$

The coordinates have the following status.

Table 2: Coordinates of the target T_ν^{osc} .

Coordinate	Meaning	Status in this paper
$D_{\text{EW}/\ell}^{\text{fixed}}$	Known electroweak and charged-lepton/neutrino source–detection substrate, including operational e , μ , and τ charged-current tags.	Fixed anchor imported from the realized-physical-interior and Standard Model results.
$\mathcal{B}_{\text{CC}}$	Boundary instruments and the operational flavor partition induced by charged-current production and detection.	Fixed interface; not a primitive global flavor ontology.
$\mathcal{P}_{\text{rec}}$	Record-coherent propagation comparison class over declared baseline, energy, path, and environment protocols; channel-level dynamics may be unitary or nonunitary.	Fixed comparison condition; “coherent” refers to common record/continuation typing, not to fundamental unitary coherence.
$\mathcal{E}_{\text{osc}}$	Empirical evidence set: nontrivial transition, baseline/energy-dependent modulation, and at least two nuisance-separated reactor response-mode classes under the explicit phenomenological response assumptions.	Empirical input, not a mathematical theorem.
$\mathcal{Q}_{\text{phys}}$	Lawful equivalence identifying simultaneous basis, phase, labeling, and representative changes that preserve the complete transition interface.	Defined in section 4.
$\mathcal{C}_{\text{open}}$	Completion coordinate containing the physical occupant or occupants of any residual role.	Deliberately left open.

field content, mass character, state-space dimension, and mechanism remain open

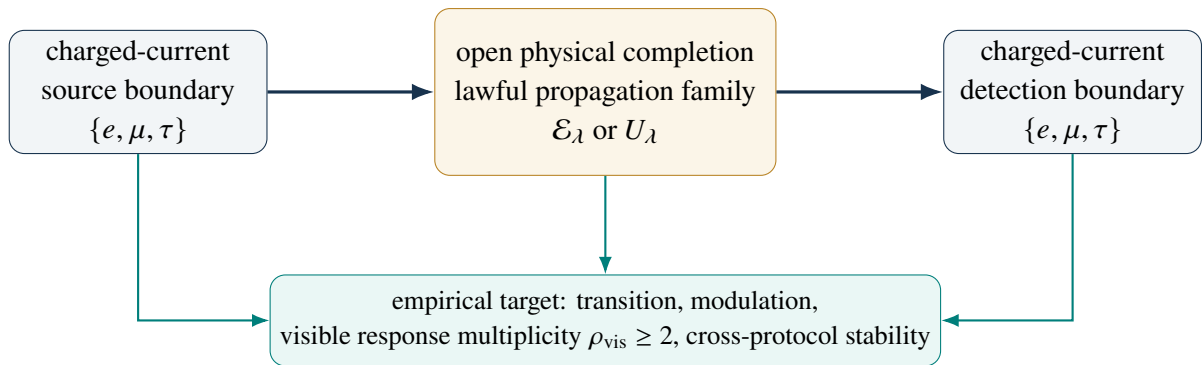


Figure 2: The source–propagation–detection target. The physical completion remains unspecified.

2.2 Fixed anchors and open completion coordinates

The target fixes a four-dimensional local relativistic quantum regime in the low-energy electroweak/lepton sector, charged-current source and detector channels, and one record-coherent boundary-to-boundary propagation interface. Record coherence means that all admitted protocols are compared through one common channel/record semantics; it does not require the microscopic dynamics to be unitary or perfectly coherent. At least the three known active boundary labels e , μ , and τ are fixed operationally, but the propagation Hilbert space is not assumed to be exactly three-dimensional and the boundary labels are not assumed to exhaust every possible state.

The empirical target includes a common reactor $\bar{\nu}_e$ survival subinterface with at least two source-supported response-mode classes. The decisive evidence comes from the long-baseline KamLAND and kilometer-baseline Daya Bay spectral analyses because the two experiments share the same charged-current electron-antineutrino boundary semantics while resolving sharply separated phase-frequency coordinates in their published response models [3, 4, 6, 7]. Solar and atmospheric observations independently establish the broader flavor-transformation target [25, 28, 29]. The phenomenological response-model assumptions are treated explicitly as empirical-analysis input and are not identified with a microscopic carrier model.

By contrast, the following remain open: field content; local or nonlocal carrier type; Dirac, Majorana, pseudo-Dirac, effective, nonunitary, Lorentz-violating, topological, or mixed character; state-space dimension; faithful global form and defect completion; numerical masses, gaps, angles, phases, and interaction strengths; mass ordering; and absolute scale.

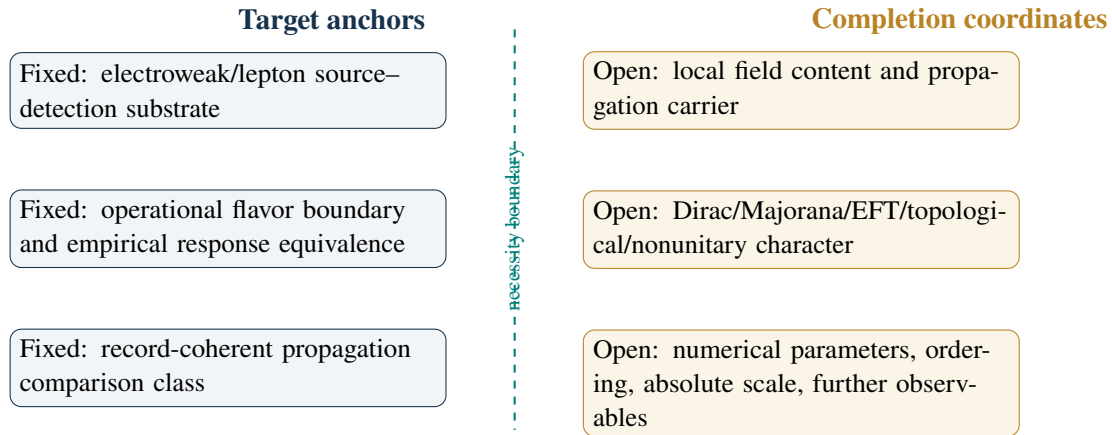


Figure 3: Fixed empirical and operational target with open completion coordinates.

2.3 Primary experimental propositions

The experimental inputs are intentionally qualitative at the theorem level. Numerical best-fit values and confidence contours support the response separation, but they are not treated as microscopic structural premises.

Definition 2.1 (Operational boundary flavor tagging). *The admitted source and detection protocols distinguish charged-current electron-, muon-, and where available tau-associated channels at the operational boundary. The labels are fixed by the electroweak interaction and detector response, not by an arbitrary basis choice in the propagation space.*

Definition 2.2 (Nontrivial flavor transition). *After declared source, detector, background, and calibration controls, at least one boundary preparation channel has nonzero probability of being reconstructed in a different active flavor channel after propagation, or an equivalent appearance record establishes the same cross-channel fact.*

The SNO neutral-current/charged-current comparison measured a non-electron active component of the solar flux significantly above zero, directly establishing flavor transformation [25]. Atmospheric and accelerator appearance analyses provide independent support, including tau-appearance evidence [24, 27]. The proof below requires only one admitted cross-channel transition record.

Definition 2.3 (Propagation-dependent modulation). *The admitted transition or disappearance kernel depends reproducibly and nontrivially on baseline, energy, path, or a controlled combination of those variables. The response is not exhausted by a static relabeling or a common phase.*

Super-Kamiokande observed a zenith-angle-dependent atmospheric muon-neutrino deficit and later an oscillatory dip in the reconstructed L/E distribution [28, 29]. KamLAND rejected an undistorted reactor-antineutrino spectrum and measured a reproducible spectral deformation [6, 7]. Daya Bay observed near/far electron-antineutrino disappearance and subsequently measured its spectral frequency [3, 4].

Definition 2.4 (Nuisance-separated reactor response modes). *On the common operational reactor $\bar{\nu}_e \rightarrow \bar{\nu}_e$ boundary, the admitted phenomenological response description contains at least two source-supported phase-frequency intervals that remain inequivalent after each collaboration's declared nuisance treatment and the common unit/calibration equivalence of theorem 2.10. In the notation of theorem 2.12,*

$$\rho_{\text{vis}}(C_\nu) \geq 2.$$

This is an empirical-model-conditioned boundary statement, not a microscopic spectral assertion.

KamLAND reports a slow reactor oscillation phase-frequency coordinate, with best fit $7.58 \times 10^{-5} \text{ eV}^2$ and separate statistical and systematic uncertainties, while Daya Bay reports a much faster direct reactor spectral-frequency coordinate, $2.59 \times 10^{-3} \text{ eV}^2$, in its stated three-neutrino response analysis [4, 6]. The numerical symbols are retained only as the common phase-frequency coordinate used by the source analyses. Their microscopic mass interpretation is not imported into the role definition.

Definition 2.5 (Cross-protocol mode stability). *The two admitted reactor mode records remain distinct under the published source/detector, normalization, energy-response, baseline-distribution, and environment nuisance treatments. A lawful calibration transport may move a mode only inside the source-supported nuisance envelope; it may not introduce an order-of-magnitude rescaling of the physical L/E coordinate or alter the response-model class after the fact.*

The separation is large compared with the reported uncertainty envelopes: the published central phase-frequency coordinates differ by a factor of about 34, whereas the quoted frequency and detector-response uncertainties are at the percent-to-few-percent scale [4, 6]. The theorem below uses only the fact that the source-supported intervals remain disjoint under the declared nuisance treatment.

Definition 2.6 (Empirical inputs do not select a completion). *The empirical evidence may instantiate the target, establish visible response modes, and falsify a proposed realization. It does not select an ungenerated completion candidate or convert a likelihood ranking into role occupancy.*

2.4 Response records and finite visible-mode multiplicity

Definition 2.7 (Calibrated response record). *A calibrated response record is a tuple*

$$R = (\Lambda_R, \mathcal{B}_R, \widehat{K}_R, N_R, M_R),$$

where Λ_R is the protocol domain, $\mathcal{B}_R$ is the operational boundary instrument, $\widehat{K}_R$ is the reconstructed transition/disappearance profile, N_R is the declared normalization and calibration information, and M_R is the controlled matter/environment information.

Definition 2.8 (Lawful response equivalence). *Two records R and R' are lawfully response-equivalent, written $R \sim_{\text{resp}} R'$, when there exists an anchor-preserving protocol map and calibration/environment transport that preserves boundary-channel semantics and carries the complete calibrated response profile of R to that of R' within the declared source uncertainty. Arbitrary rescaling of L/E , source-interaction changes, detector-flavor retyping, field-content changes, or undeclared environment terms are not equivalences.*

The quotient

$$\text{Resp}(C_\nu) := \{R \text{ admitted by } C_\nu\} / \sim_{\text{resp}}$$

is useful for comparing response records, but it does not by itself define r_Δ .

Definition 2.9 (Phenomenological reactor response datum). *For each admitted reactor protocol j , the empirical input contains a source-declared oscillatory response object*

$$\mathfrak{m}_j = (\Lambda_j, \mathcal{T}_j, \mathcal{L}_j, I_j, \Theta_j),$$

where Λ_j is the calibrated protocol domain, $\mathcal{T}_j$ is the collaboration's published survival-response model used for the spectral-frequency extraction, $\mathcal{L}_j(\xi, \vartheta)$ is the collaboration-level likelihood or χ^2 profile for a positive phase-frequency coordinate ξ and nuisance coordinates $\vartheta \in \Theta_j$, and $I_j \subset \mathbb{R}_{>0}$ is the source-supported confidence interval obtained after the stated profiling or propagation of Θ_j . The analysis map

$$\Phi_j : (\mathcal{T}_j, \mathcal{L}_j, \Theta_j) \mapsto I_j$$

is part of the target specification. The coordinate ξ is treated only as a phenomenological phase-frequency coordinate. No microscopic mass, propagation-eigenstate count, or completion mechanism is inferred from it. KamLAND's principal fit is explicitly two-flavor with $\theta_{13} = 0$ while the paper reports the same Δm_{21}^2 result in a three-neutrino treatment; Daya Bay's extraction is explicitly three-neutrino. Those model choices are therefore retained as empirical-analysis metadata rather than identified with the open microscopic completion.

Definition 2.10 (Lawful mode equivalence). *Let $\mathfrak{m}_i$ and $\mathfrak{m}_j$ be source or candidate mode records on the same charged-current boundary. They are lawfully mode-equivalent, written*

$$\mathfrak{m}_i \simeq_{\text{mode}} \mathfrak{m}_j,$$

iff, after common unit conversion, there exists a finite chain of source-authorized calibration, baseline, environment, or analysis-coordinate transports whose stated nuisance support maps the phase-frequency interval of one record into an interval overlapping the other while preserving the charged-current boundary semantics. A change of phenomenological fit model may enter such a chain only when the source analysis itself supplies the corresponding frequency-coordinate transport or stability statement; a model label by itself neither identifies nor separates two modes. The equivalence relation is the transitive closure of these lawful identifications. A finite uncertainty interval is one mode object; its individual frequency points are not separately counted.

Definition 2.11 (Boundary-visible mode set). *For a completion X , transport its event-level prediction through the fixed source, detector, baseline, and environment data and then apply the same analysis maps Φ_j used in the target specification. If the induced profiles cannot reproduce the admitted response data within the stated assumptions, X fails the target before $\rho_{\text{vis}}(X)$ is assigned. Otherwise let*

$$\mathfrak{M}_{\text{vis}}(X) := \{\text{nontrivial protocol-resolved mode records induced by } X\} / \simeq_{\text{mode}}.$$

The target mode set $\mathfrak{M}_{\text{vis}}(C_\nu)$ is defined from the admitted experimental response data. This construction is

operational and target-relative: a single nonlinear microscopic law may induce several mode classes, while several microscopic parameters may induce one.

Definition 2.12 (Boundary-visible response multiplicity). *Define*

$$\rho_{\text{vis}}(X) := |\mathfrak{M}_{\text{vis}}(X)| \quad (2)$$

when the visible mode set is finite, and $\rho_{\text{vis}}(X) = \infty$ otherwise. For a finite mode set this is equivalently $\dim_{\mathbb{C}} \mathbb{C}[\mathfrak{M}_{\text{vis}}(X)]$; the free-vector-space form is only a formal encoding of cardinality and carries no physical superposition claim. A nonzero-width source uncertainty interval contributes one mode object, not uncountably many frequency characters.

The empirical assumption required by the response theorem is therefore explicit:

$$\rho_{\text{vis}}(C_{\nu}) \geq 2. \quad (3)$$

KamLAND's published slow phase-frequency interval and Daya Bay's published fast interval remain disjoint after their stated profiling, nuisance treatment, and common unit conversion [4, 6]. KamLAND obtains its principal interval from a two-flavor fit and reports stability of the extracted slow frequency in a three-neutrino treatment, while Daya Bay explicitly obtains its fast frequency in a three-neutrino framework. The analysis maps and model metadata are therefore included in the empirical assumptions rather than treated as model-independent facts extracted directly from raw counts.

Proposition 2.13 (Empirical mode-class separation). *Under theorems 2.4 and 2.5, the target's common reactor survival interface contains at least two inequivalent boundary-visible response-mode classes:*

$$|\mathfrak{M}_{\text{vis}}(C_{\nu})| = \rho_{\text{vis}}(C_{\nu}) \geq 2.$$

Proof. KamLAND's high-statistics analysis rejects an undistorted reactor spectrum at greater than 5σ and reports a best-fit slow phase-frequency coordinate $7.58 \times 10^{-5} \text{ eV}^2$ with quoted statistical and systematic uncertainties [6]. Daya Bay's near/far spectral analysis reports a direct fast phase-frequency coordinate $2.59 \times 10^{-3} \text{ eV}^2$ in its stated three-neutrino response framework [4]. The central coordinates differ by about a factor of 34, while the source-supported uncertainty envelopes are small compared with that separation; Daya Bay also reports alternative positron energy-response models consistent at roughly the percent level. Hence no source-declared calibration, nuisance, or analysis-coordinate transport identifies the two reported intervals. The different fit-model labels are not used as the separator; the disjoint source-supported phase-frequency intervals are. They therefore determine two distinct classes of $\simeq_{\text{mode}}$.

The proposition is explicitly conditional on the stated phenomenological response assumptions. It does not claim that the two classes are model-independent Fourier atoms of the raw event spectra. A future common-boundary analysis that invalidated those assumptions or represented both experimental responses by one lawful mode class would reopen theorem 2.4 and the r_{Δ} branch. $\square$

2.5 Target identity and controlled variants

Definition 2.14 (Target fingerprint). *The fingerprint of T_{ν}^{osc} is the ordered tuple*

$$\mathfrak{f}(T_{\nu}^{\text{osc}}) = (D_{\text{EW}/\ell}^{\text{fixed}}, \mathcal{B}_{\text{CC}}, \mathcal{P}_{\text{rec}}, \sim_{\text{resp}}, \mathfrak{P}_e^{\text{phen}}, \{\Phi_j\}_j, \mathfrak{M}_{\text{vis}}(C_{\nu}), \simeq_{\text{mode}}, \mathcal{E}_{\text{osc}}, C_{\text{open}}).$$

A comparison is same-target only if this tuple is preserved up to the lawful equivalence fixed in section 4.

Theorem 2.15 (Stability of the fixed target). *No later step in the necessity argument may change a coordinate of $\mathfrak{f}(T_{\nu}^{\text{osc}})$ to rescue a failed conclusion. Adding a completion candidate is allowed only through*

the predeclared open coordinate C_{open} ; changing a target anchor requires a new target specification and a fresh role analysis.

Proof. The target fingerprint individuates the object whose role necessities are being tested. A change of boundary instruments, environment policy, response-mode assumptions, lawful mode equivalence, or record-coherent comparison class changes either well-formedness or empirical content. Under fixed-domain reference and no-repair, the changed object is a successor target, not a repaired instance of T_v^{osc} . $\square$

Controlled variants include an explicitly matter-dependent solar propagation target, open-system propagation, sterile-extended state space, nonstandard source/detection interactions, and curved or dynamical spacetime. These variants are not denied; they are treated separately because they alter one or more target coordinates. The general channel theorem in section 6 covers a broad class of open propagation maps without silently importing source/detection changes.

3 Dependency structure and no-backflow condition

The argument uses established AASC and Standard Model results at sharply delimited points. The distinction between imported mathematical structure, locally proved neutrino-specific statements, empirical input, and completion-specific ideas is essential: no downstream representation is permitted to supply the conditions that make the upstream role analysis possible.

3.1 Imported mathematical and physical results

Table 3: Imported theorem families and their use in the present argument.

ID	Source	Positive use	Not inferred
KERNEL	Non-Degenerate Construction and the Kernel of Admissibility	Forced {Adm, St, Ref, Irr} kernel, fixed-domain identity, anti-weakening, and no same-act repair [15, Thm. 2.34, Thm. 6.20].	No neutrino-specific role, empirical record, or carrier.
AMETRIC/ UNUS	AMetric Boundary; Unus Solus Possibilis Est	No boundary selector or graded admission; uniqueness of the admissible interior once the target and existence conditions are independently fixed [18, Thm. 5.26, Thm. 14.2]; [22, Thm. 6, Thm. 7].	No physical existence theorem and no choice among completions.
RPI	The Realized Physical Interior	Fields, localized particle classes, bounded transport, deformation response, vacuum, quantum continuation, and record quotient [19, Thm. 13.1, 14.1, 16.1, 20.1, 26.1, 38.2].	No occupancy of the neutrino-specific roles proved here.

ID	Source	Positive use	Not inferred
ROC	Role-Occupancy Closure	Forced-role-set discipline, candidate/realization distinction, and the later compatible-assignment criterion [16, Thm. 4.1, Thm. 6.1, Thm. 7.4].	The present paper does not invoke the sufficiency direction to realize a carrier.
SR1	Simultaneous Constraint Collapse	The warning that independently admissible projections must still be compatible through one physical witness structure [17, Thm. 7.3, Thm. 8.2, Thm. 9.4, Thm. 10.1].	No candidate elimination is performed here.
SYM-5	Flavor Symmetry, Basis Skin, Spurion Structure	Minimal massless-neutrino no-PMNS guard and neutrino-target bifurcation [12, Sec. 8.1, Thm. SYM5-T17].	No numerical mixing coordinate.
SYM-7	Accidental Symmetry by Operator Dimension	Separation of minimal, Dirac, renormalizable-Majorana, and Weinberg-EFT fibers [11, Thm. 5.1].	No preferred fiber.
SM-5	Flavor Transition, Mixing, and Generation	PMNS non-primitivity, oscillation-route status, and boundary-reconstruction classification [13, Thm. 7.2, Thm. 11.1, Thm. 11.2].	No fitted PMNS value or standard oscillation formula as an upstream premise.
SM-6/SYM-11	Standard Model carrier and symmetry closure	The minimal Standard Model is closed at its declared scope; new field/operator content is a target extension [20, 21].	No claim that all physical extensions are excluded.

3.2 No-backflow theorem

Definition 3.1 (Backflow). *A downstream object backflows when its successful representation of a target is used to authorize an earlier role, target anchor, or admissibility condition that was required before the representation could be stated.*

Theorem 3.2 (No-backflow condition). *None of the following may serve as a premise in the role-necessity chain: a numerical PMNS matrix, the standard mass-eigenstate oscillation formula, a chosen Dirac or Majorana realization, an AASC neutrino-ordering or numerical-readout result, a higher-form or non-invertible symmetry completion, or a candidate new particle. Such an object may be used only after the target-forced role set is fixed, as a comparator, consistency check, or later candidate.*

Proof. Each listed object contains completion-specific structure absent from the target fingerprint. PMNS and mass-eigenstate formulae represent one familiar realization; Dirac/Majorana labels select distinct target fibers; numerical AASC neutrino results occur downstream of additional standing and calibration assumptions; generalized-symmetry data can constrain an occupant only after the relevant physical interface

has been established; and a new particle is itself a candidate carrier. Using any of them to define the upstream role would make the role conclusion conditional on the object it is meant to constrain. This violates the forced-role ordering imported from Role-Occupancy Closure and the target identity of theorem 2.14. $\square$

3.3 Empirical and mathematical inputs occupy different proof positions

The AASC sources establish constructional, physical-role, target-separation, and classification results. They do not independently establish that neutrino flavor transformation occurs. The empirical inputs are therefore grounded in the primary experiments cited in section 2 and documented in section B. Repetition of a conclusion across theoretical papers is not a substitute for experimental evidence.

Conversely, the experiments do not establish the AASC kernel, Role-Occupancy Closure, or the distinction between a role family and a target-specific role instance. The two source classes enter different parts of the proof.

Proposition 3.3 (Import-scope preservation). *Every imported theorem used here retains its original target, hypotheses, and nonclaim boundary. A local statement requiring a stronger reading than the cited source is not inherited from that source.*

Proof. The domain, hypotheses, and conclusion of an imported theorem define its admissible use. Strengthening any of them changes the theorem object. The source-provenance appendix records the principal imports used in the argument. $\square$

3.4 Excluded completion-specific inputs

The following are intentionally absent from the load-bearing necessity proof:

- global neutrino-fit tables and current best-fit numerical coordinates;
- AASC neutrino numerical-readiness, ordering, and matter-response results;
- non-invertible or higher-form neutrino model proposals;
- seesaw, radiative, sterile, extra-dimensional, Lorentz-violating, decay, or decoherence models;
- cosmological and beta-decay mass bounds;
- any assumption of an experimentally accessible carrier scale.

These may become relevant when candidate completions are analyzed. Their exclusion here is a dependency-order requirement, not a judgment about their physical merit.

Dependency summary

The locally proved neutrino-specific chain begins with the empirical target and the carrier-neutral interface of section 4. AASC results constrain target identity and role typing; they do not supply the observed transition or the completion that realizes it.

4 Framework-neutral source–propagation–detection interface

4.1 Why an effective interface is sufficient for necessity

A complete quantum-field-theoretic treatment of neutrino production, propagation, and detection includes localized source and detector states, wave packets, finite coherence, external particles, and an amplitude that joins the two interaction regions. Standard QFT analyses recover an effective oscillation kernel in the experimental regimes of interest [1, 2, 5, 8]. The present analysis does not derive that reduction anew.

It isolates a structural consequence shared by every description that admits a well-defined boundary-to-boundary transition kernel: if the propagation map preserves each operational flavor sector, it cannot produce an admitted cross-flavor transition.

To make the scope maximally clear, the paper formulates the interface first for quantum channels and then gives the unitary representative used for algebraic examples and countermodels. No unitary-coherence assumption is built into the target definition.

4.2 Boundary and propagation objects

Definition 4.1 (Record-relevant state space). $\mathcal{H}_\nu$ is a separable Hilbert space sufficient to represent the record-relevant propagation degrees of freedom for the fixed target. It may contain more than the three active boundary sectors. No fundamental field content or exact state-space dimension is assumed.

Definition 4.2 (Operational boundary algebra). Let P_e, P_μ, P_τ be mutually orthogonal projections representing the idealized charged-current flavor readout on the active boundary support

$$P_A := P_e + P_\mu + P_\tau \leq \mathbf{1}_{\mathcal{H}_\nu}.$$

The operational boundary algebra is

$$\mathcal{A}_{\text{CC}} := \text{alg}\{P_e, P_\mu, P_\tau, P_A\}.$$

It records source/detection channel semantics. It is not assumed to be a complete algebra of propagation observables.

Definition 4.3 (Boundary preparation). An ideal preparation in channel α is a density operator ρ_α satisfying

$$\rho_\alpha = P_\alpha \rho_\alpha P_\alpha, \quad \text{Tr } \rho_\alpha = 1.$$

Realistic source impurity and detector confusion are carried by the empirical calibration data and may be represented by positive-operator-valued effects. The exact theorem below is stated for the ideal quotient record after those corrections.

Definition 4.4 (Propagation channel family). For every declared protocol $\lambda = (L, E, \gamma, \mathcal{M}, \dots)$, let

$$\mathcal{E}_\lambda : \mathcal{T}_1(\mathcal{H}_\nu) \longrightarrow \mathcal{T}_1(\mathcal{H}_\nu)$$

be a normal completely positive trace-preserving map on trace-class operators. Its Heisenberg adjoint is $\mathcal{E}_\lambda^\dagger : \mathcal{B}(\mathcal{H}_\nu) \rightarrow \mathcal{B}(\mathcal{H}_\nu)$. The family may represent a unitary coherent evolution, an isometric dilation with unobserved sectors, or a controlled open-system variant. Source and detector interactions remain fixed by $\mathcal{B}_{\text{CC}}$.

The use of completely positive maps is standard for general quantum state changes [9, 26]. No claim is made that an arbitrary channel is a fundamental neutrino dynamics; the channel language is used only to state the necessary boundary-transfer result without assuming unitary three-state propagation.

Definition 4.5 (Transition kernel). The ideal boundary-to-boundary transition kernel is

$$K_{\beta\alpha}(\lambda) := \text{Tr}[P_\beta \mathcal{E}_\lambda(\rho_\alpha)]. \quad (4)$$

For a unitary representative $\mathcal{E}_\lambda(\rho) = U_\lambda \rho U_\lambda^\dagger$,

$$K_{\beta\alpha}(\lambda) = \text{Tr}\left(P_\beta U_\lambda \rho_\alpha U_\lambda^\dagger\right). \quad (5)$$

Definition 4.6 (Propagation algebra). *For a unitary representative, the propagation algebra $\mathcal{A}_{\text{prop}}$ is the von Neumann algebra generated by the record-relevant family $\{U_\lambda\}$. Its commutant is $\mathcal{A}'_{\text{prop}}$. The condition $\mathcal{A}_{\text{CC}} \subseteq \mathcal{A}'_{\text{prop}}$ is the unitary aligned case.*

4.3 Lawful redescription

A simultaneous unitary change of representation V acts by

$$P_\alpha \mapsto VP_\alpha V^\dagger, \quad \rho_\alpha \mapsto V\rho_\alpha V^\dagger, \quad \mathcal{E}_\lambda \mapsto \text{Ad}_V \circ \mathcal{E}_\lambda \circ \text{Ad}_{V^\dagger}. \quad (6)$$

It changes no boundary-to-boundary probability.

Theorem 4.7 (Lawful-redescription invariance). *Under equation (6), every $K_{\beta\alpha}(\lambda)$ is invariant. Boundary-face invariance under $\mathcal{E}_\lambda$, pointwise invariance of $\mathcal{A}_{\text{CC}}$ under $\mathcal{E}_\lambda^\dagger$, the unitary commutant relation $\mathcal{A}_{\text{CC}} \subseteq \mathcal{A}'_{\text{prop}}$, the calibrated mode-class set $\mathfrak{M}_{\text{vis}}$, and the visible response multiplicity ρ_{vis} are also invariant.*

Proof. For the kernel,

$$\begin{aligned} K'_{\beta\alpha}(\lambda) &= \text{Tr}[VP_\beta V^\dagger V\mathcal{E}_\lambda(V^\dagger V\rho_\alpha V^\dagger V)V^\dagger] \\ &= \text{Tr}[VP_\beta \mathcal{E}_\lambda(\rho_\alpha)V^\dagger] = K_{\beta\alpha}(\lambda) \end{aligned}$$

by cyclicity of the trace. Conjugation carries fixed-point equations and commutators to conjugate equations and commutators, preserving whether they vanish. The visible spectrum is extracted from the complete calibrated kernel on the fixed L/E coordinate after quotienting only declared nuisance directions. Simultaneous representation change leaves that kernel and calibration data unchanged, so it cannot create, remove, merge, or split visible response modes and therefore preserves ρ_{vis} . $\square$

Corollary 4.8 (No chart-level physical role). *A weak basis, propagation eigenbasis, row/column phase convention, eigenvector ordering, or PMNS parameterization is not by itself standing-bearing content of the interface.*

4.4 Physical quotient

Definition 4.9 (Physical quotient). $\mathcal{Q}_{\text{phys}}$ identifies two interface presentations when they are connected by a finite chain of transformations preserving

- (i) the complete transition kernel $K_{\beta\alpha}(\lambda)$;
- (ii) the operational meaning of the boundary channels;
- (iii) the declared matter/environment data;
- (iv) the response equivalence $\sim_{\text{resp}}$, phenomenological response-model anchor, lawful mode equivalence, and visible mode set $\mathfrak{M}_{\text{vis}}$;
- (v) the target fingerprint $\mathfrak{f}(T_v^{\text{osc}})$.

A transformation that changes only one component while holding the others fixed is not automatically lawful.

This quotient is intentionally stricter than matrix similarity alone. A unitary matrix can be a skin change only when the source states, detector effects, and propagation map transform together. Rotating a propagation operator while leaving the charged-current boundary instruments fixed changes the incidence relation and can change the physics.

4.5 Minimal assumptions and scope

The core theorem uses the following assumptions:

- Q1. The boundary-to-boundary record is represented by a well-defined channel family $\mathcal{E}_\lambda$.
- Q2. The idealized flavor sectors are mutually exclusive at the quotient level after declared source/detector corrections.
- Q3. One physical completion is responsible for the complete admitted record set and its visible response-mode set; unrelated theories may not be assigned to different experiments without an explicit target split.
- Q4. Lawful redescription preserves the entire kernel and response quotient.
- Q5. Matter and environment effects are declared in λ or in a controlled target variant.
- Q6. The empirical records are imported qualitatively; fitted numerical coordinates are not used as structural premises.

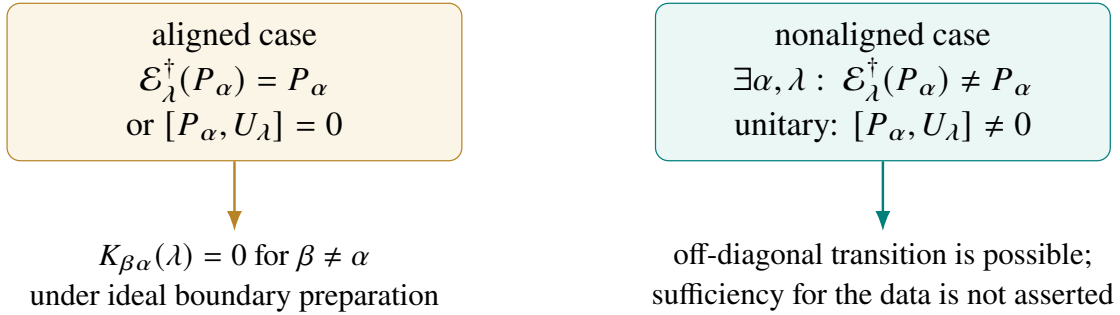


Figure 4: Aligned and nonaligned boundary/propagation structure. The right-hand condition is necessary but not sufficient for the full oscillation data.

Representative boundary

The finite-dimensional unitary formula is a controlled representative, not a claim that neutrinos are literally prepared as exact flavor projectors or propagate as plane waves. The channel theorem in section 6 states the necessary result directly at the effective source–detector kernel level. A QFT model that yields the same kernel lies in the same comparison class; one that changes the operational boundary or response quotient defines a target variant.

5 Carrier-neutral target roles

5.1 Roles are defined by physical work

A target-specific role is not a synonym for the familiar object that normally realizes it. Two distinct pieces of physical work are required by the fixed interface: failure of charged-current boundary-face invariance under the complete propagation channel, and boundary-visible response multiplicity of at least two on the common reactor survival interface. The first is an incidence/transport role; the second is a response-structure role. Neither is defined by a mass matrix, PMNS chart, or microscopic carrier count.

Definition 5.1 (Well-typed target role). *A target role is well typed when it is specified by:*

- (i) *a fixed target operation or invariant;*
- (ii) *a lawful-equivalence class;*
- (iii) *a same-target deletion condition;*

- (iv) a positive witness type;
- (v) an occupancy question distinct from the role definition; and
- (vi) an explicit list of carrier descriptions that are not part of the definition.

5.2 Interaction–propagation nonalignment

For each active boundary projector define the boundary face

$$\mathfrak{F}_\alpha := \{\rho \geq 0 : \text{Tr } \rho = 1, \rho = P_\alpha \rho P_\alpha\}.$$

Definition 5.2 (The role $r_{\bowtie}$). *The interaction–propagation nonalignment role $r_{\bowtie}$ is occupied when the complete record-relevant propagation/interface family fails to preserve at least one operational charged-current boundary face:*

$$\exists \alpha, \lambda : \mathcal{E}_\lambda(\mathfrak{F}_\alpha) \not\subseteq \mathfrak{F}_\alpha. \quad (7)$$

Equivalently, some state prepared in one charged-current boundary sector acquires support outside that sector. In a unitary representative this is equivalent, on the record-relevant support, to

$$\exists \alpha, \lambda : [P_\alpha, U_\lambda] \neq 0,$$

and hence to failure of the aligned commutant condition $\mathcal{A}_{\text{CC}} \subseteq \mathcal{A}'_{\text{prop}}$.

The role is a support-transport/incidence relation, not a matrix. A standard PMNS representative can display it, but so can an enlarged state space, a nonunitary channel, an environment-coupled propagation law, or another lawful completion.

Deletion. Delete $r_{\bowtie}$ by imposing boundary-face invariance $\mathcal{E}_\lambda(\mathfrak{F}_\alpha) \subseteq \mathfrak{F}_\alpha$ for every α, λ while holding the source states, detector effects, target records, and comparison class fixed. Certified cross-boundary transition is then impossible; the nontrivial flavor-transition target is lost.

Positive witness. A lawful occupant supplies a quotient-stable boundary-to-propagation support-transfer relation. It need not supply a preferred basis, a finite-dimensional matrix, or a local particle.

5.3 Boundary-visible response multiplicity

Definition 5.3 (The role r_Δ). *The propagation-response differentiation role r_Δ is the target-specific obligation that one lawful physical completion possess boundary-visible response multiplicity*

$$\rho_{\text{vis}}(X) \geq \rho_{\text{vis}}(C_\nu) \geq 2 \quad (8)$$

on the fixed common reactor electron-antineutrino survival interface, while preserving the target boundary, calibration, environment, and lawful quotient.

The notation Δ is mnemonic for differentiation, not a commitment to a mass-squared difference. The role is carried by the multiplicity of lawful, nuisance-separated boundary response-mode classes defined in section 2. For a finite mode set the equivalent quantity $\dim \mathbb{C}[\mathfrak{M}_{\text{vis}}(X)]$ is merely a formal encoding of that multiplicity. One microscopic law may induce several visible mode classes; many microscopic parameters may induce one. The present theorem constrains only the operational interface projection.

Deletion. Delete r_Δ by replacing the same-boundary response map with a one-mode-class response while preserving $r_{\bowtie}$, the charged-current boundary, phenomenological response assumptions, target calibration, nuisance/environment data, and one-completion identity. This is a physical alteration of

the continuation/response map, not deletion or merger of a target record. A one-class candidate cannot injectively realize a target with at least two nuisance-separated mode classes.

Positive witness. A lawful occupant must supply a single physical interface whose projected reactor survival response realizes at least the two target mode classes under the same protocol-level extraction and nuisance rules. It may realize them through one mechanism, several coupled mechanisms, a continuum, a nonunitary channel, or another carrier type.

5.4 No definitional equivalence to mass and mixing

Proposition 5.4 (No mass/PMNS definitional equivalence). *Neither $r_{\bowtie}$ nor r_{Δ} is definitionally equivalent to a neutrino mass term, a nondegenerate mass spectrum, a PMNS matrix, or a numerical mixing parameter.*

Proof. $r_{\bowtie}$ is defined by boundary-face transport under the full channel family. Any physical structure that moves support between operational boundary sectors can occupy it. The unitary commutator is a representative, not the definition.

r_{Δ} is defined by the finite multiplicity $|\mathfrak{M}_{\text{vis}}(X)|$ of lawful response-mode classes. Distinct experimental responses contribute distinct classes only when they remain inequivalent under the stated calibration/nuisance relation; repeated measurements of the same mode collapse to one class. Their microscopic origin is left open. Therefore neither microscopic parameter count nor mass-spectrum cardinality is definitionally equivalent to the role. $\square$

5.5 Upstream roles and target-specific role instances

Role family	Upstream status	Disposition here
Field carrier Φ_D	Occupied	Not a residual role.
Localized particle class Π_D	Occupied for active boundary channels	Not a residual role.
Quantum predictive continuation	Occupied	Supplies general channel/state language.
Bounded transport and causal ordering	Occupied at role level	Not a neutrino-specific propagation completion.
Generic deformation/mass response M_D	Occupied as a role family	Does not instantiate the target-specific response multiplicity.
Record-fixation quotient	Occupied	Used to admit the empirical propositions of section 2.
$r_{\bowtie}$	Not decided upstream	Target-specific channel-incidence role proved below.
r_{Δ}	Not decided upstream	Target-specific response-multiplicity role proved below.

Table 4: Upstream roles and target-specific coordinates.

5.6 Eligibility, occupancy, and realization

Role definition, role eligibility, occupancy, and realization are different relations. This paper establishes the role definitions and tests one known comparison fiber for nonoccupancy; it does not construct the general eligibility relation or a complete occupancy fiber.

5.7 The nested role complex

A one-mode-class nonaligned model can transfer support and display one visible modulation mode. Therefore nonalignment alone does not imply r_{Δ} . Conversely, every target realization with $\rho_{\text{vis}} \geq 2$ contains

at least two admitted nontrivial response-mode classes on the charged-current boundary and therefore cannot preserve every boundary face. The exact proof appears in section 8.

Definition 5.5 (Nested role complex). *The target-specific role complex is*

$$\mathcal{R}_v^{\text{osc}} := (r_{\triangleright\triangleleft}, r_{\triangle}; r_{\triangle} \succ r_{\triangleright\triangleleft}), \quad (9)$$

where $r_{\triangle} \succ r_{\triangleright\triangleleft}$ records $r_{\triangle} \Rightarrow r_{\triangleright\triangleleft}$ while $r_{\triangleright\triangleleft} \not\Rightarrow r_{\triangle}$.

Remark 5.6 (No carrier cardinality follows). *The ordered pair in (9) does not imply two distinct physical occupants. One candidate structure may perform both pieces of work. Carrier cardinality can be addressed only after candidate decomposition and compatibility are known.*

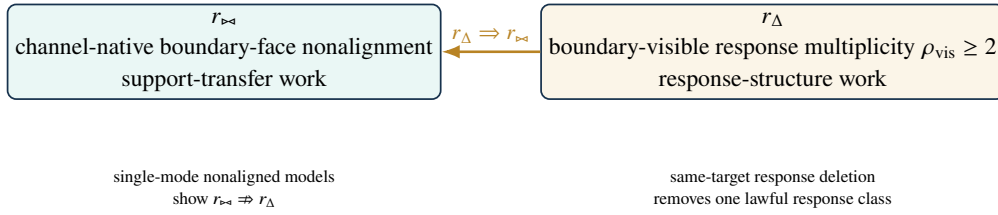


Figure 5: Nested role complex: channel-native nonalignment is the weaker requirement; visible response multiplicity of at least two is the stronger requirement.

6 Interaction–propagation nonalignment necessity

6.1 Boundary flavor is fixed before propagation

The first local burden is to show that the source/detection flavor partition can be stated independently of a propagation basis. The operational charged-current boundary does exactly this: a source or detector is tagged by the associated charged lepton. The abstract projectors P_{α} are therefore boundary instruments, not spectral projectors.

Lemma 6.1 (Boundary flavor-interface lemma). *ER-0 together with the known charged-current source/detection structure determines the boundary algebra $\mathcal{A}_{\text{CC}}$ up to lawful simultaneous redescription. It does not determine a propagation basis or PMNS representative.*

Proof. ER-0 fixes operationally distinct source/detection channels by the charged lepton observed or produced at the boundary. In the controlled representative these channels are mutually exclusive projection effects P_{α} . A simultaneous unitary conjugation of all boundary and propagation objects changes the representative but preserves every transition kernel, so $\mathcal{A}_{\text{CC}}$ is fixed only up to $\mathcal{Q}_{\text{phys}}$. No spectral property of U_{λ} is used in the construction. Hence the boundary algebra is well-defined before any propagation diagonalization or mixing chart. $\square$

6.2 Common phase cannot produce modulation

Lemma 6.2 (Common-phase no-modulation lemma). *If $U_{\lambda} = e^{i\phi(\lambda)} \mathbf{1}$ on the active record-relevant support for every λ , then*

$$K_{\beta\alpha}(\lambda) = \text{Tr}(P_{\beta}\rho_{\alpha}) = \delta_{\alpha\beta}$$

under ideal preparation. The transition kernel is independent of baseline, energy, path, and internal propagation structure and cannot realize ER-1 or ER-2.

Proof. The phase cancels in $U_{\lambda}\rho_{\alpha}U_{\lambda}^{\dagger}$. Ideal support gives $\rho_{\alpha} = P_{\alpha}\rho_{\alpha}P_{\alpha}$. For $\beta \neq \alpha$, orthogonality implies $P_{\beta}\rho_{\alpha} = 0$; for $\beta = \alpha$, normalization gives unit trace. Thus no off-diagonal transition or modulation

remains. $\square$

The lemma does not claim that every degenerate microscopic model is literally a common phase on every hidden degree of freedom. It states the target-relevant condition: if all structure visible to the boundary transition kernel collapses to a central phase, the fixed target record cannot be reproduced.

6.3 Commuting boundary sectors cannot exchange support

Lemma 6.3 (Commuting-boundary no-transition lemma). *Assume ideal preparation $\rho_\alpha = P_\alpha \rho_\alpha P_\alpha$ and mutually orthogonal boundary projectors. If*

$$[P_\gamma, U_\lambda] = 0 \quad \forall \gamma, \lambda,$$

then

$$K_{\beta\alpha}(\lambda) = 0 \quad \text{for all } \beta \neq \alpha.$$

Proof. Commutation gives

$$U_\lambda \rho_\alpha U_\lambda^\dagger = U_\lambda P_\alpha \rho_\alpha P_\alpha U_\lambda^\dagger = P_\alpha U_\lambda \rho_\alpha U_\lambda^\dagger P_\alpha.$$

Therefore the propagated state remains supported in $\text{im } P_\alpha$. For $\beta \neq \alpha$,

$$P_\beta U_\lambda \rho_\alpha U_\lambda^\dagger = P_\beta P_\alpha U_\lambda \rho_\alpha U_\lambda^\dagger P_\alpha = 0,$$

and taking the trace yields the result. $\square$

Corollary 6.4 (Transition implies nonalignment). *If ER-1 holds for the effective channel interface, then at least one charged-current boundary face is not propagation invariant. Hence $r_{\triangleright\triangleleft}$ is target-forced. In a unitary representative this implies $\mathcal{A}_{\text{CC}} \not\subseteq \mathcal{A}'_{\text{prop}}$.*

Proof. If every boundary face were invariant under every effective channel, theorem 12.1 would make every certified cross-boundary kernel vanish. This contradicts ER-1. The unitary commutant statement follows as the representative specialization. $\square$

6.4 Deletion theorem

Theorem 6.5 (Nonalignment deletion theorem). *Deleting $r_{\triangleright\triangleleft}$ while preserving the target anchors, record-coherent comparison class, and ideal boundary support eliminates cross-flavor transition and therefore destroys T_v^{osc} .*

Proof. Deleting $r_{\triangleright\triangleleft}$ means imposing invariant boundary faces $\mathcal{E}_\lambda(\mathfrak{F}_\alpha) \subseteq \mathfrak{F}_\alpha$ for all α, λ without changing the source/detection boundary, the empirical record, or the completion policy. By theorem 12.1, the resulting transition kernels have no off-diagonal support. ER-1 is an admitted component of the target evidence, so the modified object no longer realizes T_v^{osc} . A simultaneous redescription cannot restore the transition because commutation and the kernels are conjugation invariant. Therefore $r_{\triangleright\triangleleft}$ is a target-forced role. $\square$

6.5 Realistic boundary imperfections

The ideal projector proof isolates the structural statement. Real sources and detectors have backgrounds, finite resolution, misidentification, and production/detection wave packets. These do not invalidate the theorem; they alter the record admission and introduce bounded deviations from ideal support. Let

$$\epsilon_\alpha := 1 - \text{Tr}(P_\alpha \rho_\alpha)$$

measure source leakage and let E_β be the actual detector effect with

$$\|E_\beta - P_\beta\|_\infty \leq \eta_\beta.$$

Under boundary-aligned propagation, the gentle-measurement estimate yields

$$\text{Tr}(E_\beta U_\lambda \rho_\alpha U_\lambda^\dagger) \leq 2\sqrt{\epsilon_\alpha} + \eta_\beta \quad (\beta \neq \alpha), \quad (10)$$

up to the declared normalization convention. A proof is given in section C. Thus an admitted off-diagonal record exceeding the certified source/detector contamination bound still forces nonalignment. Experimental analyses already contain the required background and misidentification controls; this paper uses their admitted qualitative conclusions rather than recalculating them.

Proposition 6.6 (Robust nonalignment). *Suppose ER-1 is admitted only after the experiment-specific background and misidentification evidences establish an off-diagonal signal exceeding the bound allowed by ideal-support violations and detector cross-talk. Then the conclusion of theorem 6.4 remains valid for the corresponding effective interface.*

Proof. If propagation were aligned, (10) would bound every off-diagonal detection probability by the declared contamination envelope. The admitted ER-1 signal lies outside that envelope. Therefore alignment is inconsistent with the certified record. $\square$

6.6 Channel generalization

The central result is not peculiar to unitary dynamics. Let $\mathcal{E}_\lambda$ be a completely positive trace-preserving channel and $\mathcal{E}_\lambda^\dagger$ its Heisenberg dual. If every boundary projector is fixed,

$$\mathcal{E}_\lambda^\dagger(P_\alpha) = P_\alpha,$$

then

$$\text{Tr}(P_\beta \mathcal{E}_\lambda(\rho_\alpha)) = \text{Tr}(\mathcal{E}_\lambda^\dagger(P_\beta) \rho_\alpha) = \text{Tr}(P_\beta \rho_\alpha) = 0$$

for $\beta \neq \alpha$. More generally, invariance of each boundary block under the channel suffices. The complete statement appears in theorem 12.1. This is now the primary definition-level form of r_{ns} , not merely an extension of a unitary theorem. Decoherence or nonunitarity therefore does not by itself evade the role theorem; it either preserves every boundary face and cannot create certified transfer, or it supplies nonalignment in channel form. The target comparison class requires common record semantics, not fundamental unitary coherence.

6.7 Source/detection new physics is a target variant

A common objection is that flavor change may be produced at the source or detector rather than during propagation. That possibility is physically legitimate, but it changes $\mathcal{B}_{\text{CC}}$ or the preparation/detection maps. the known charged-current boundary is held fixed in the present target. A new source interaction, nonstandard detector effect, or flavor-changing matter interaction is therefore a candidate in a broader coupled completion or an explicit target variant. It is not a same-target counterexample obtained while preserving the boundary algebra.

The point is not semantic. If the boundary instruments change, the operational meaning of P_α , ρ_α , or E_β changes. The theorem can be reapplied to the new target, but its role incidence must be recomputed. The fixed-target condition therefore does not permit a modified source or detector boundary to be counted as a realization of the original aligned target.

6.8 What has and has not been proved

Nonalignment theorem boundary

The local theorem proves that a fixed charged-current boundary and a certified cross-flavor transition record require a propagation/interface channel that does not preserve every boundary face. It does not prove that the mismatch is unitary, three-dimensional, caused by masses, represented by a PMNS matrix, or confined to propagation rather than a declared coupled source–propagation–detection completion.

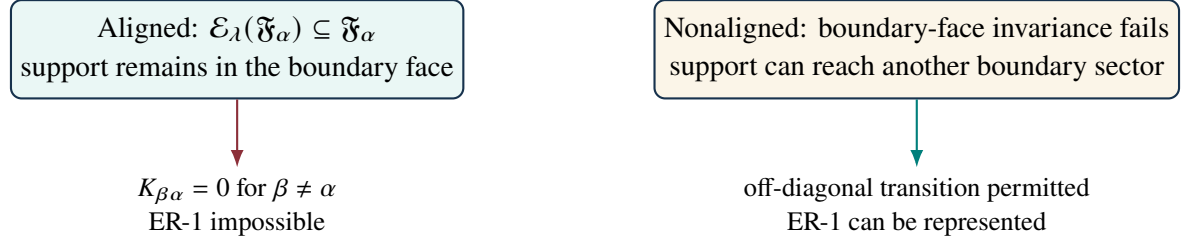


Figure 6: The nonalignment theorem is a support-transport statement, not a mass-mixing ansatz.

7 Boundary-visible propagation-response multiplicity

7.1 Finite operational mode classes

The second target-specific role is defined on a finite operational object rather than on an unspecified function-space quotient. A visible mode is a source- or candidate-supported phase-frequency record consisting of a protocol range, a declared phenomenological response model, a source-supported frequency interval, and the associated nuisance treatment. Mode equivalence is generated only by lawful calibration, baseline, environment, and analysis-coordinate transports. The resulting multiplicity counts inequivalent mode classes, not frequency points, experiments, fit parameters, microscopic eigenstates, or carriers.

7.2 Empirical response-model assumptions

The r_Δ result is explicitly conditional on the response models used by the experimental collaborations. KamLAND and Daya Bay analyze reactor $\bar{\nu}_e$ disappearance through declared oscillatory survival-response models. KamLAND reports a slow phase-frequency coordinate near $7.58 \times 10^{-5} \text{ eV}^2$ with separate statistical and systematic uncertainties, while Daya Bay reports a much faster direct spectral-frequency coordinate near $2.59 \times 10^{-3} \text{ eV}^2$ in a stated three-neutrino response framework [4, 6]. These quantities enter only as phenomenological boundary-response coordinates together with their published likelihood or χ^2 profiles, confidence intervals, and nuisance treatment.

The argument is therefore not a model-independent reconstruction from raw event counts. If a later common-boundary analysis invalidated the admitted response-model class, or showed that the two experimental responses descend to one lawful mode class, the r_Δ conclusion would have to be reconsidered.

7.3 Empirical mode-class admission

Proposition 7.1 (Empirical mode-class admission). *Under theorems 2.4 and 2.5, the common calibrated reactor survival interface satisfies*

$$\rho_{\text{vis}}(C_\nu) = |\mathfrak{M}_{\text{vis}}(C_\nu)| \geq 2.$$

Proof. By theorems 2.9 and 2.10, each collaboration contributes a finite mode record obtained from its published response analysis after the stated nuisance treatment. KamLAND’s high-statistics spectrum rejects an undistorted spectrum at greater than 5σ and reports a slow frequency coordinate with quoted

statistical and systematic uncertainties [6]. Daya Bay’s near/far spectral analysis reports a fast frequency coordinate in its declared three-neutrino response framework [4]. Their central values differ by a factor of about 34, and the source-supported intervals are disjoint by more than an order of magnitude. Daya Bay additionally reports alternative positron energy-response models mutually consistent at roughly the percent level in the analysis supporting the frequency extraction [4].

A lawful mode equivalence must arise from source-authorized nuisance, calibration, environment, or analysis-coordinate transports together with common unit conversion. The fact that the two experiments use different phenomenological fit formulations does not itself separate the modes; the separation is the failure of every authorized transport to move the slow interval into overlap with the fast interval. The two response records therefore define distinct classes in $\mathfrak{M}_{\text{vis}}(C_\nu)$. $\square$

Remark 7.2 (Interpretive boundary). *The proposition does not claim that raw reactor data contain two model-independent Fourier atoms. It does not infer two microscopic masses, two fields, two particles, or two fundamental Hamiltonian eigenvalue gaps. It states only that, under the explicitly stated phenomenological response assumptions, the common reactor boundary contains at least two nuisance-separated response-mode classes.*

7.4 Physical mode projection of a completion

Let X be an arbitrary completion. Its event-level reactor prediction is first transported through the fixed source, detector, baseline, and environment data. The same analysis maps Φ_j used in the target specification are then applied protocol by protocol, with the same nuisance and calibration semantics. If X cannot reproduce the admitted response data for a protocol, it fails the target before $\rho_{\text{vis}}(X)$ is assigned. Otherwise its induced visible mode set is the finite or countable quotient $\mathfrak{M}_{\text{vis}}(X)$ of theorem 2.11.

For finite $\mathfrak{M}_{\text{vis}}(X)$,

$$\rho_{\text{vis}}(X) = |\mathfrak{M}_{\text{vis}}(X)| = \dim_{\mathbb{C}} \mathbb{C}[\mathfrak{M}_{\text{vis}}(X)].$$

The free-vector-space expression is only a formal representation of finite multiplicity and has no physical-superposition significance. In particular, a confidence interval is one mode object; its continuum of internal frequency values is not counted as a continuum of physical modes.

Definition 7.3 (Response-multiplicity occupancy). *A completion X occupies r_Δ iff it supplies one lawful source–propagation–detection interface such that the target mode classes inject into its induced visible mode set,*

$$\iota_X : \mathfrak{M}_{\text{vis}}(C_\nu) \hookrightarrow \mathfrak{M}_{\text{vis}}(X),$$

with each image class reproducing the corresponding calibrated boundary response under the fixed protocol and nuisance assumptions. In particular,

$$\rho_{\text{vis}}(X) \geq \rho_{\text{vis}}(C_\nu) \geq 2.$$

The injection condition is stronger than bare cardinality: the completion must reproduce the correct nuisance-separated response classes. A single nonlinear microscopic law may induce both target classes; several coupled structures may induce them jointly; a continuum may induce additional classes. No microscopic carrier cardinality follows.

7.5 Boundary-visible response-multiplicity theorem

Theorem 7.4 (Boundary-visible response-multiplicity theorem). *Conditional on the phenomenological response assumptions of section 2 and on theorems 2.4 and 2.5, every realization of T_ν^{osc} must occupy r_Δ .*

Proof. A realization of T_ν^{osc} must reproduce every target-individuating physical coordinate. By theorem 7.1, the common reactor target contains at least two distinct classes in $\mathfrak{M}_{\text{vis}}(C_\nu)$. By definition of realization,

the completion's protocol-level projection must reproduce the calibrated response associated with each of those classes. Hence there must be an injective map ι_X from the target classes into $\mathfrak{M}_{\text{vis}}(X)$, and therefore $\rho_{\text{vis}}(X) \geq 2$.

This conclusion is not obtained by declaring two experiments to be two dimensions. Two measurements of the same lawful response mode would be identified by $\simeq_{\text{mode}}$ and contribute only one class. The present target contributes at least two because the two source-supported frequency intervals remain separated under the admitted calibration and nuisance transformations. $\square$

7.6 Same-target response-multiplicity deletion

Theorem 7.5 (Response-multiplicity deletion theorem). *Let X be a completion occupying r_Δ . Construct a same-target candidate $X^{(1)}$ by changing only the physical propagation-response map so that, after the same source/detector/environment projection and the same analysis maps,*

$$|\mathfrak{M}_{\text{vis}}(X^{(1)})| = 1,$$

while preserving the charged-current boundary, phenomenological response assumptions, calibration/nuisance data, nonalignment $r_{\text{p}\Delta}$, and the one-completion identity. Then $X^{(1)}$ does not realize T_{ν}^{osc} .

Proof. The target mode set contains at least two inequivalent classes. A one-class candidate admits no injection $\mathfrak{M}_{\text{vis}}(C_{\nu}) \hookrightarrow \mathfrak{M}_{\text{vis}}(X^{(1)})$ and therefore cannot reproduce both target response classes under the same analysis and nuisance rules. The target itself has not been modified: the failure lies in the candidate's continuation/response map. Since $r_{\text{p}\Delta}$ is preserved, the deletion isolates physical work beyond mere boundary support transfer. $\square$

7.7 Visible multiplicity is not microscopic multiplicity

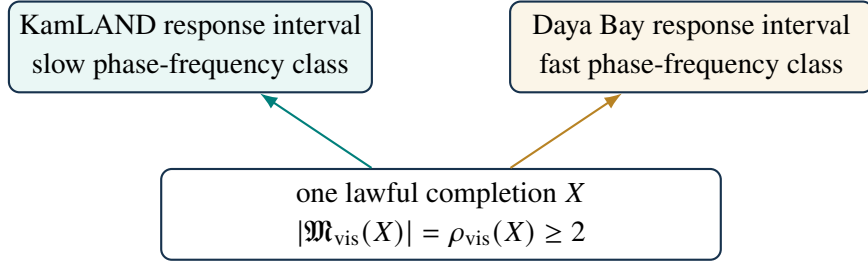
No inference of the form

$$\rho_{\text{vis}} = 2 \implies \text{two masses, two gaps, two fields, two particles, or two couplings}$$

is licensed. ρ_{vis} counts target-relative, nuisance-separated response-mode classes. One nonlinear law may generate several such classes; many microscopic parameters may project to one; an open or continuous microscopic spectrum may reproduce the same two-class boundary response. The role therefore remains carrier-neutral.

7.8 Response multiplicity and nonalignment

If every charged-current boundary face is invariant under the full propagation channel, an ideal electron-antineutrino source state remains in the electron boundary face and the ideal survival response is protocol-trivial apart from the fixed contamination and calibration data. Such a channel cannot induce either admitted nontrivial reactor mode class. Hence $r_\Delta \implies r_{\text{p}\Delta}$. The strictness of that implication is established in section 8 by an explicit nonaligned single-mode model.



Each finite experimental uncertainty interval is one mode object. The multiplicity counts lawful response classes, not experiments, frequency points, or microscopic parameters.

Figure 7: Boundary-visible response-multiplicity requirement on the common reactor electron-antineutrino interface.

Scope of the response theorem

The response-multiplicity result is conditional on the collaboration-level phenomenological response analyses described above. It does not re-fit raw events, claim a model-independent Fourier decomposition, identify the phase-frequency coordinates with microscopic carrier data, or claim that the target is exhausted by exactly two modes. A valid common-boundary analysis that reduced the two admitted responses to one lawful mode class would reopen r_{Δ} .

8 Role noncollapse, nested implication, and anti-skin closure

8.1 The role-equivalence question

The two target-specific coordinates now carry manifestly different physical work. $r_{\bowtie}$ is boundary-face transport/incidence. r_{Δ} is the multiplicity of lawful response-mode classes visible on the fixed reactor survival boundary under the stated phenomenological response assumptions. Role implication without converse implication does not collapse these tasks into one role.

Definition 8.1 (Role equivalence). *Two target-specific roles r and s are role-equivalent on T_v^{osc} when, for every lawful completion in the declared comparison class, occupancy of r is equivalent to occupancy of s , their same-target deletion failures coincide, and their positive witness sets descend to the same $\mathcal{Q}_{\text{phys}}$ -class. Role implication without the reverse implication is not role equivalence.*

8.2 A single-mode nonaligned model

Let $\mathcal{H}_v = \mathbb{C}^2$, let

$$P_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

and let

$$U_t = e^{-it\omega\sigma_x} = \cos(\omega t)\mathbf{1} - i\sin(\omega t)\sigma_x, \quad \omega \neq 0.$$

For $\rho_0 = P_0$,

$$K_{10}(t) = \sin^2(\omega t).$$

The channel moves support between boundary faces and therefore occupies $r_{\bowtie}$. Under the fixed protocol-level response analysis, this model presents one nontrivial oscillatory phase scale and therefore one visible mode class, so $\mathfrak{M}_{\text{vis}} = \{[\mathfrak{m}_{\omega}]\}$ and $\rho_{\text{vis}} = 1$.

Proposition 8.2 (Single-mode nonalignment countermodel). *There exists a well-formed carrier-neutral interface that occupies $r_{\bowtie}$ but not r_{Δ} .*

Proof. The model above is nonaligned for generic t and produces cross-boundary transition. Its calibrated response induces a single lawful mode class under the source response-model extraction, hence $\rho_{\text{vis}} = 1$. Therefore $r_{\triangleright\Delta}$ holds while the multiplicity-two condition defining r_{Δ} fails. $\square$

8.3 Hidden spectral richness is not automatically visible response multiplicity

Consider

$$\mathcal{H}_v = \mathbb{C}^2 \otimes \mathbb{C}^m, \quad P_0 = |0\rangle\langle 0| \otimes \mathbf{1}_m, \quad P_1 = |1\rangle\langle 1| \otimes \mathbf{1}_m,$$

with a block-diagonal family

$$U_t = P_0 V_0(t) + P_1 V_1(t),$$

where V_0 and V_1 may have arbitrarily rich internal spectra. Every boundary face remains invariant. Internal phases that never alter the protocol-extracted charged-current response produce no element of $\mathfrak{M}_{\text{vis}}$.

Proposition 8.3 (Internal-richness anti-promotion). *Microscopic spectral multiplicity, hidden-sector phase multiplicity, or internal channel complexity does not occupy r_{Δ} unless it descends to at least two inequivalent classes in the boundary-visible mode set.*

Proof. The displayed family preserves the boundary faces, so cross-boundary kernels vanish. Any internal spectral distinction invisible to the admitted boundary response produces no protocol-resolved mode class and contributes no element to $\mathfrak{M}_{\text{vis}}$. Visible response multiplicity is therefore not microscopic spectral multiplicity. $\square$

8.4 Visible response multiplicity entails nonalignment

Theorem 8.4 (Response-multiplicity-to-nonalignment implication). *Within the fixed comparison class,*

$$r_{\Delta} \implies r_{\triangleright\Delta}.$$

Proof. Suppose $r_{\triangleright\Delta}$ were absent. Then every charged-current boundary face would be invariant under every effective propagation channel. In particular an ideal state prepared in the electron boundary face remains in that face. The calibrated reactor survival probability is then unity up to the fixed source/detector contamination data and cannot produce any nontrivial source-supported response-mode class. Thus $\mathfrak{M}_{\text{vis}} = \emptyset$ and $\rho_{\text{vis}} = 0$, contradicting occupancy of r_{Δ} , which requires $\rho_{\text{vis}} \geq 2$. $\square$

Theorem 8.5 (Strict nested role theorem). *The target-specific role coordinates are non-equivalent and strictly nested:*

$$r_{\Delta} \succ r_{\triangleright\Delta},$$

meaning $r_{\Delta} \implies r_{\triangleright\Delta}$ while $r_{\triangleright\Delta} \not\Rightarrow r_{\Delta}$.

Proof. The forward implication is theorem 8.4. The reverse implication fails by theorem 8.2. Their deletion tests are also different: deleting $r_{\triangleright\Delta}$ removes boundary support transfer entirely, whereas deleting r_{Δ} can leave nonalignment and one visible modulation mode intact while reducing the candidate to one lawful response-mode class. Therefore the roles are not merely two names for the same target requirement. $\square$

Remark 8.6 (Why the response role is not mere record coverage). *The second role is defined on the candidate's physical response map rather than by counting experimental records. Its deletion holds the target fixed, preserves nonalignment, and reduces the candidate from at least two lawful response classes to one. The distinction is therefore physical-response structure versus boundary incidence, not data-set count versus boundary incidence.*

8.5 Weak basis, rephasing, and chart closure

Theorem 8.7 (Anti-skin and no-basis-generation theorem). *No simultaneous lawful weak-basis transformation, rephasing, propagation-eigenvector relabeling, or PMNS-chart change can create, destroy, or change $r_{\triangleright\triangleleft}$, $\mathfrak{M}_{\text{vis}}$, or ρ_{vis} .*

Proof. By theorem 4.7, simultaneous lawful conjugation preserves all boundary-to-boundary kernels and boundary-face invariance. The visible mode set is extracted from the complete calibrated physical response through the fixed protocol maps, so a representation change that preserves that response leaves its mode classes unchanged. A transformation applied only to propagation while the charged-current boundary is held fixed is a physical candidate variation, not a lawful redescription. $\square$

Corollary 8.8 (No PMNS-from-skin inference). *A nontrivial PMNS representative can display occupancy of the role complex in a conventional completion, but it cannot authorize either role from below and no numerical PMNS coordinate follows from this paper.*

8.6 Role matrix and target failures

Interface class	$r_{\triangleright\triangleleft}$	r_{Δ}	Target disposition
Aligned common phase	No	No	ER-1 and ER-2 fail.
Aligned with hidden spectral richness	No	No	Internal modes are invisible to the charged-current response.
Nonaligned single-mode model	Yes	No	Flavor transfer possible; ER-3 two-mode-class burden fails.
One lawful completion with $\rho_{\text{vis}} \geq 2$	Yes	Yes	Meets the target-specific role burden; carrier identity remains open.
Patchwork of unrelated one-mode models	Possibly per model	No joint occupancy	Violates the one-completion anchor or requires a target split.

Table 5: Role occupancy and target disposition after the response-multiplicity construction.

Role-complex conclusion

The exact target forces two non-equivalent target coordinates with the order $r_{\Delta} \succ r_{\triangleright\triangleleft}$. The stronger coordinate is a physical response-multiplicity invariant, not a second carrier slot. One future candidate may occupy both, several compatible candidates may jointly occupy them, or no admissible completion may do so.

9 Occupancy in the minimal Standard Model comparison fiber

9.1 Why occupancy is tested only after the role analysis

The preceding sections fixed the target and derived its target-specific role complex without inspecting the completion inventory. Only at this point is the known physical substrate tested for occupancy. This order prevents a residual role from being defined merely as “whatever the minimal Standard Model lacks.” The roles follow from the empirical target, the carrier-neutral interface, and same-target deletion consequences; occupancy is a separate comparison.

Let

$$R_{\text{upstream}} = \{\Phi_D, \Pi_D, \text{BoundTrans}_D, G_D, M_D, \Omega_D, T_D, \text{Cau}_D, \text{Record}_D, \text{QuantumCont}_D, \dots\}$$

denote the relevant upstream role families imported from the realized physical interior. Let $D_{\text{SM}}^{\nu 0}$ denote the minimal renormalizable Standard Model fiber with its observed charged-current lepton architecture, no right-handed neutrino, and no dimension-five Weinberg operator.

9.2 Upstream role discharge

Theorem 9.1 (Upstream-role discharge). *On the known physical substrate, every general role needed to formulate the source–propagation–detection problem is already occupied at its imported scope:*

$$R_{\text{upstream}} \subseteq R_{\text{occupied}}^{\text{known}}.$$

In particular, the present analysis does not introduce fields, particles, quantum continuation, localized source/detection, bounded propagation, gauge organization, generic deformation response, vacuum, time, causality, or record fixation as new neutrino roles.

Proof. The realized physical interior supplies fields as distributed persistence carriers, particles as localized persistence classes, bounded transport, gauge equivalence, generic deformation/mass response, vacuum baseline, temporal order, causal dependence, and record-fixation quotienting [19, Thms. 13.1, 14.1, 16.1, 17.1, 18.1, 19.1, 20.1, 26.1]. The Standard Model structural substrate specializes these roles to an electroweak lepton sector with charged-current source and detector channels [20, 21]. The quantum predictive projection supplies lawful state evolution and measurement records at the general role level. None of these results supplies the target-specific propagation incidence or boundary-visible response multiplicity; they discharge only the upstream role families. $\square$

9.3 The minimal neutrino fiber

The minimal renormalizable Standard Model contains left-handed neutrino components inside the lepton doublets but no gauge-singlet right-handed neutrinos. With one Higgs doublet and only operators of dimension at most four, no gauge-invariant neutrino mass operator exists. The dimension-five Weinberg operator lies outside this fiber [30]. Consequently the active neutrinos are massless in the declared comparison fiber and the neutrino kinetic/gauge sector has an enhanced flavor-redescription freedom. SYM-5 and SM-5 use this fact to block a primitive PMNS assignment: without a neutrino mass or another standing propagation tensor, a putative lepton mixing matrix can be removed by a neutrino weak-basis transformation [12, 13].

Choose the charged-lepton mass basis for display. On the active neutrino support, vacuum propagation in the minimal fiber is proportional to the identity up to a common kinematic phase:

$$U_{\text{vac}}^{\nu 0}(L, E) = e^{i\phi(L, E)} \mathbf{1}_A.$$

Ordinary coherent forward scattering in matter adds the standard charged-current potential in the electron boundary sector and neutral-current terms proportional to the identity. In this massless comparison fiber the matter Hamiltonian remains diagonal in the same charged-current flavor decomposition:

$$H_{\text{mat}}^{\nu 0} = \text{diag}(V_e, 0, 0) + c \mathbf{1}_A, \quad [H_{\text{mat}}^{\nu 0}, P_\alpha] = 0.$$

It can create relative diagonal phases but cannot transport support between flavor sectors. Thus neither vacuum nor standard matter propagation occupies r_{∞} .

Lemma 9.2 (Minimal-fiber propagation fingerprint). *Within $D_{\text{SM}}^{\nu 0}$ and the fixed charged-current boundary,*

$$\mathcal{A}_{\text{CC}} \subseteq (\mathcal{A}_{\text{prop}}^{\nu 0})', \quad K_{\beta\alpha}^{\nu 0}(\lambda) = 0 \quad (\beta \neq \alpha),$$

and the boundary-visible response-mode set is empty, so $\rho_{\text{vis}} = 0$.

Proof. The vacuum generator is central on the active support. The standard matter term is a linear combination of the boundary projectors and the identity. Both commute with $\mathcal{A}_{CC}$, as does their path-ordered evolution when the matter profile varies along the trajectory, because all instantaneous generators remain diagonal in the same mutually orthogonal projectors. The channel therefore preserves every boundary block. The no-transition theorem gives the vanishing of off-diagonal kernels. Without transition or path-dependent survival modulation, no nontrivial protocol-resolved mode class is induced, so $\mathfrak{M}_{\text{vis}} = \emptyset$ and $\rho_{\text{vis}} = 0$. $\square$

9.4 No PMNS occupancy in the massless fiber

Proposition 9.3 (PMNS nonoccupancy in the massless fiber). *A nontrivial numerical PMNS matrix does not have standing as a target-specific propagation occupant in $D_{\text{SM}}^{\nu 0}$.*

Proof. With a completely degenerate massless neutrino sector and no additional propagation tensor, an arbitrary unitary rotation among active neutrino fields leaves the kinetic and gauge terms invariant. Any matrix written in the charged-current vertex can be absorbed into that neutrino field rotation while the charged-lepton mass basis is fixed. The complete transition kernel remains diagonal and unchanged. Hence the matrix is weak-basis skin rather than a standing propagation mismatch in this fiber. This is the no-PMNS guard proved in the imported flavor/mixing sources at their declared scope [12, 13]. $\square$

9.5 Generic mass response is not the target-specific response instance

A possible objection is that the realized-physical-interior theorem already forces a generic mass/deformation-response role, so r_{Δ} is occupied before the neutrino-specific analysis begins. This confuses a role family with a target-specific instance.

Proposition 9.4 (No generic-role transfer). *The upstream existence of the generic deformation-response family M_D does not occupy r_{Δ} or $r_{\bowtie}$ in $D_{\text{SM}}^{\nu 0}$.*

Proof. M_D states that localized persistence classes in a realized physical interior require some admissible deformation-response profile. It neither supplies a neutrino-specific operator nor fixes nonalignment of the charged-current boundary with propagation. A target-specific occupancy claim must exhibit the relevant interface witness. By theorem 9.2, the minimal neutrino fiber has no such witness. Transferring generic role standing to a distinct target instance without an independent target gate would violate fixed-domain and Role-Occupancy Closure discipline [15, 16]. $\square$

9.6 No charged-lepton or gauge-sector substitution

Charged-lepton Yukawa structure distinguishes e , μ , and τ , and electroweak interactions use those labels operationally. That structure supplies the boundary algebra $\mathcal{A}_{CC}$; it does not supply a propagation map that moves neutrino support between the boundary sectors. Likewise, gauge and anomaly closure make the source/detection architecture consistent but do not generate a neutrino propagation mismatch. Boundary typing is a prerequisite for $r_{\bowtie}$, not its occupant.

Lemma 9.5 (Boundary/occupant separation). *The physical structures that individuate the charged-current boundary channels cannot, by that fact alone, occupy either target-specific propagation role.*

Proof. If boundary individuation alone occupied $r_{\bowtie}$, every theory with distinguishable source channels would exhibit cross-channel propagation. The aligned models of theorems 6.3 and 8.3 are counterexamples. If it occupied r_{Δ} , the number of charged-lepton labels would determine the number of lawful visible response-mode classes, contradicted by the single-mode nonaligned model theorem 8.2. The roles therefore require propagation-interface witnesses beyond boundary individuation. $\square$

9.7 Radiative, nonperturbative, and hidden-sector possibilities

Within the declared renormalizable fiber, perturbative Standard Model corrections do not generate a neutrino mass operator absent an allowed operator and carrier. Electroweak nonperturbative effects violate $B + L$ in multi-fermion processes but do not supply the bilinear propagation structure required here. A gravitationally generated Weinberg term, an unobserved singlet sector, an anomaly-inflow completion, or any other additional structure changes the comparison fiber and belongs to the open family of section 11. Such possibilities may later occupy the roles; they cannot be counted as already present in $D_{\text{SM}}^{\nu 0}$.

Theorem 9.6 (Minimal-fiber target-specific nonoccupancy). *For the minimal renormalizable Standard Model comparison fiber,*

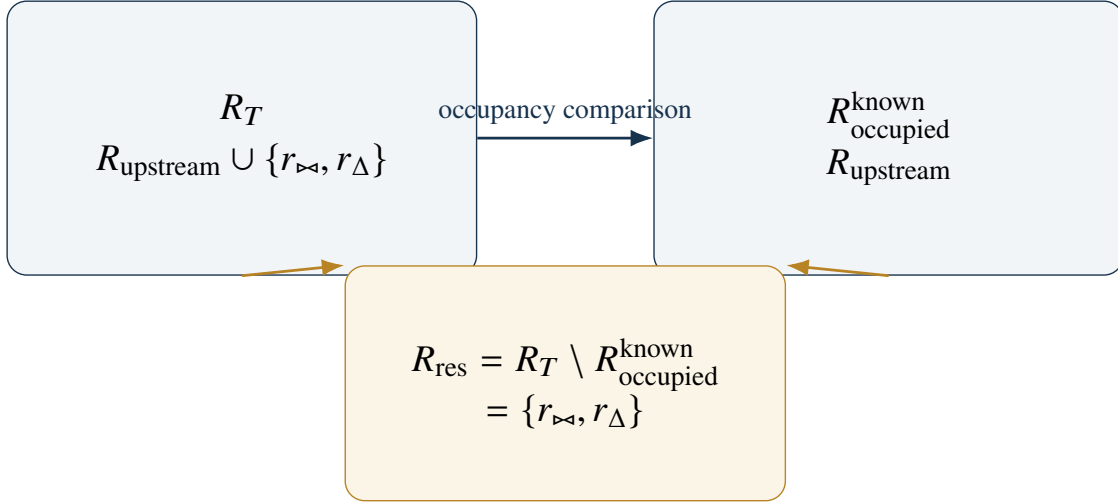
$$r_{\triangleright\triangleleft} \notin R_{\text{occupied}}^{\text{known}}(D_{\text{SM}}^{\nu 0}), \quad r_{\Delta} \notin R_{\text{occupied}}^{\text{known}}(D_{\text{SM}}^{\nu 0}).$$

Equivalently, the known substrate supplies the operational boundary and all upstream role families but not the target-specific role complex required by T_{ν}^{osc} .

Proof. Nonoccupancy of $r_{\triangleright\triangleleft}$ follows from theorem 9.2. Nonoccupancy of r_{Δ} follows because the minimal fiber has $\mathfrak{M}_{\text{vis}} = \emptyset$ and $\rho_{\text{vis}} = 0$, whereas the target requires $\rho_{\text{vis}} \geq 2$ by theorem 7.1. Theorems 9.3 and 9.4 excludes the principal presentation and role-transfer pseudo-occupants. Additions that alter the target are not occupants of the original fiber. $\square$

Required work	Known-substrate status	Basis
Charged-current source/detection channels	Occupied	Known electroweak lepton architecture.
Field/particle/transport/record roles	Occupied	Realized-physical-interior and Standard Model imports.
Generic deformation-response family	Occupied at role-family level	M_D ; no target-specific transfer.
Interaction–propagation nonalignment $r_{\triangleright\triangleleft}$	Unoccupied	$\mathcal{A}_{\text{CC}} \subseteq (\mathcal{A}_{\text{prop}}^{\nu 0})'$.
Boundary-visible response multiplicity r_{Δ}	Unoccupied	No transition kernel; $\mathfrak{M}_{\text{vis}} = \emptyset$ and $\rho_{\text{vis}} = 0$.
Mass mechanism, PMNS, ordering, scale	Not target roles here	Explicitly open completion coordinates.

Table 6: Occupancy of the relevant roles in the minimal Standard Model comparison fiber.



role obligation only: no carrier identity, cardinality, or existence theorem

Figure 8: The upstream physical role architecture is occupied; the target-specific propagation role complex is not.

10 Residual physical-role obligation

10.1 Ordered role complex

Because the boundary-visible response-multiplicity role r_{Δ} strictly implies the channel-native nonalignment role $r_{\triangleright\Delta}$, the result should not be interpreted as two independently occupiable boxes. The correct object is an ordered role complex,

$$\mathfrak{R}_{\nu}^{\text{osc}} = (\{r_{\triangleright\Delta}, r_{\Delta}\}, r_{\Delta} \succ r_{\triangleright\Delta}).$$

Its carrier cardinality is undetermined. One occupant may realize the complete complex; several compatible occupants may divide the work; or no admissible completion may survive.

Let

$$R_T(C_{\nu}, T_{\nu}^{\text{osc}}) = R_{\text{upstream}} \cup \{r_{\triangleright\Delta}, r_{\Delta}\}$$

be the target role set, typed by the precedence relation, and let

$$R_{\text{occupied}}^{\text{known}}(C_{\nu}, D_{\text{SM}}^{\nu 0}) = R_{\text{upstream}}$$

be the roles occupied in the minimal Standard Model comparison fiber.

Theorem 10.1 (Forced target-role composition). *Conditional on the empirical propositions of section 2 and the interface assumptions of section 4,*

$$R_T(C_{\nu}, T_{\nu}^{\text{osc}}) = R_{\text{upstream}} \cup \{r_{\triangleright\Delta}, r_{\Delta}\}, \quad r_{\Delta} \succ r_{\triangleright\Delta}.$$

No additional target-specific role is established by the present argument.

Proof. The upstream roles are the imported prerequisites discharged in theorem 9.1. Nontrivial flavor transformation forces $r_{\triangleright\Delta}$ by theorem 6.5. The common reactor evidence yields $\rho_{\text{vis}}(C_{\nu}) \geq 2$ by theorem 7.1, and the response theorem and its same-target deletion force r_{Δ} by theorems 7.4 and 7.5. Their strict relation is theorem 8.5. Source/detection individuation is already contained in the upstream role structure, while mass mechanism, field content, dimensionality, global form, and numerical parameters are completion coordinates rather than target roles. Hence the displayed complex is the complete target-specific addition

proved at this scope. □

10.2 Residual set difference

Definition 10.2 (Residual role complex). *The residual role content relative to the minimal comparison fiber is*

$$R_{\text{res}}(C_\nu, T_\nu^{\text{osc}}; D_{\text{SM}}^{\nu 0}) := R_T(C_\nu, T_\nu^{\text{osc}}) \setminus R_{\text{occupied}}^{\text{known}}(C_\nu, D_{\text{SM}}^{\nu 0}),$$

with the precedence and lawful-equivalence data retained.

Theorem 10.3 (Residual role-obligation theorem). *Under the target specification, empirical assumptions, interface conditions, and imported-scope constraints stated above,*

$$R_{\text{res}}(C_\nu, T_\nu^{\text{osc}}; D_{\text{SM}}^{\nu 0}) = (\{r_{\triangleright\triangleleft}, r_\Delta\}, r_\Delta \succ r_{\triangleright\triangleleft}) \neq \emptyset. \quad (11)$$

Proof. By theorem 10.1, the target role complex is $R_{\text{upstream}} \cup \{r_{\triangleright\triangleleft}, r_\Delta\}$. By theorems 9.1 and 9.6, the minimal comparison fiber occupies the relevant upstream roles and neither target-specific coordinate. The set difference therefore leaves $\{r_{\triangleright\triangleleft}, r_\Delta\}$, and theorem 8.5 supplies the retained precedence relation. Because the roles were defined before the occupancy comparison, the result is not inferred from the known absence of a neutrino mass operator. □

10.3 No carrier identity follows

Corollary 10.4 (No carrier identity follows). *Theorem 10.3 does not determine whether the residual role complex is occupied by existing but differently organized Standard Model structure in a richer fiber, new local matter, composite or solitonic structure, a topological or defect sector, a nonunitary channel, modified source/detection structure in a target variant, or another completion class. It also does not determine the number of occupants.*

Proof. The present paper does not construct a complete candidate universe, role-eligibility relation, joint compatibility predicate, or lawful assignment quotient. These are required before the sufficiency direction of Role-Occupancy Closure can license realization [16, Def. 7.1, Thms. 7.3–7.4]. The residual theorem concerns unoccupied physical work relative to one comparison fiber, not the identity of what performs that work. □

Logical boundary

The conclusion that the minimal massless-neutrino Standard Model does not occupy the target-specific role complex is not the conclusion that a new particle exists. The latter would require a complete candidate analysis and a separate realization theorem.

10.4 Failure loci

A same-target challenge to equation (11) must reopen at least one of three loci.

Theorem 10.5 (Residual-endpoint failure trichotomy). *Any same-scope defeat of theorem 10.3 must establish at least one of:*

- (i) **target reopening**: *an empirical proposition, target anchor, or record-equivalence clause is false, incomplete, or improperly model-loaded;*
- (ii) **role reopening**: *a deletion theorem fails, the role complex collapses further, or a lawful redescription removes the asserted work;*
- (iii) **occupancy reopening**: *the minimal Standard Model comparison fiber contains a target-specific occupant omitted by the analysis.*

A carrier proposal outside the comparison fiber does not defeat the theorem; it belongs to a distinct completion fiber.

Proof. The endpoint is the set difference of a target role complex and an occupied-role set. A same-scope counterexample must therefore change the former, the latter, or the lawful identity under which the difference is taken. These are exactly the three cases listed above. Adding a new fiber changes the comparison object and cannot establish occupancy in the original fiber. $\square$

10.5 Constraint-induced character of the result

The result is generative only in the constraint-residue sense: a realized target forces a physical work obligation, and the declared known substrate is shown not to discharge it. No object is manufactured by the constraint argument. This is structurally analogous to the generation-cardinality precedent, in which the role load is fixed before the open candidate coordinate is closed [14].

Residual endpoint

The realized neutrino oscillation interface requires a nonempty target-specific residual role complex. The residual is physical because deleting it destroys the fixed target, and carrier-neutral because no occupant has been selected or inferred.

11 Controlled family of completion targets

11.1 Why an open completion family is required

The residual theorem is evaluated against the minimal massless-neutrino Standard Model fiber, but possible realizations of the target may lie in richer physical theories. A right-handed-neutrino theory, a renormalizable Majorana theory, and a Weinberg-EFT description cannot be treated as mere redescrptions of the minimal Standard Model: they differ in field content, operator algebra, EFT order, symmetry status, and continuation structure. SYM-5 and SYM-7 explicitly separate these targets [11, 12].

The appropriate comparison is therefore a controlled family with common empirical and operational anchors and distinct completion fibers.

Definition 11.1 (Common neutrino base). *Let*

$$B_\nu = (D_{\text{EW}/\ell}^{\text{known}}, \mathcal{B}_{\text{CC}}, \mathcal{E}_{\text{osc}}, \mathcal{Q}_{\text{phys}}, \mathcal{P}_{\text{record}})$$

collect the common known electroweak/lepton substrate, charged-current boundary, admitted oscillation records, lawful physical quotient, and record protocol. It contains no neutrino mass operator, extra field, or completion mechanism.

Definition 11.2 (Open completion family). *The neutrino completion family is the typed disjoint union*

$$\mathfrak{T}_\nu = \coprod_{\lambda \in \Lambda_{\text{comp}}} T_{\nu, \lambda}, \quad T_{\nu, \lambda} = (B_\nu, C_\lambda),$$

where C_λ records the fiber-specific fields, operators, symmetries, global data, scales, and continuation laws. The coproduct symbol emphasizes target distinction; it is not a claim that all fibers coexist in one theory.

11.2 Four benchmark fibers

The imported symmetry analysis supplies four useful benchmark fibers.

ID	Field/operator content	Characteristic status	Target distinction
NU0	No ν_R ; no dimension-five operator	Massless active neutrinos in the renormalizable SM	Comparison fiber used in the occupancy analysis.
NUD	Gauge-singlet ν_R plus Dirac Yukawa operator	Dirac mass route; enlarged field and flavor structure	New local carrier and operator content.
NURM	ν_R , Dirac Yukawa, and Majorana mass M_R	Renormalizable Majorana/seesaw-capable theory	Lepton-number, field, and heavy-scale structure differ from NUD.
NUW	Dimension-five Weinberg operator in the low-energy EFT	Majorana mass route without explicit low-energy ν_R	EFT order, cutoff/matching, and operator content differ from renormalizable fibers.

Table 7: Controlled benchmark completion fibers.

Theorem 11.3 (Well-typing of the open completion family). *The family $\mathfrak{T}_\nu$ permits the completion coordinates to remain open without identifying target-inequivalent fibers. Candidates may be compared across fibers only through explicitly typed matching, projection, or empirical-equivalence maps; no such map transports standing or role occupancy by default.*

Proof. All fibers share B_ν by construction, so they address the same empirical interface at the level declared common. Their C_λ entries differ in target-individuating data. Fixed-domain discipline therefore forbids treating those differences as same-scope skin. A matching map can relate low-energy observables, but unless it preserves the complete field/operator, symmetry, boundary, and continuation invariant bundle, it is not target identity. The transfer theorem of the AASC kernel and the explicit four-fiber separation in SYM-7 block automatic standing transport [11, 15]. $\square$

Corollary 11.4 (No same-target repair by added structure). *Adding ν_R , M_R , the Weinberg operator, a new interaction, or a topological sector after NU0 fails the oscillation target creates a new candidate fiber. It does not convert NU0 itself into a realization of the target.*

11.3 Matching is not identity

A high-scale NURM model can match onto a low-energy NUW EFT. This relation is physically important, but it is not identity of theory objects. The high-energy fiber contains heavy fields and renormalizable operators absent from the EFT; the EFT contains matching coefficients and a validity envelope. Any later candidate analysis must represent the matching arrow and test whether the complete source–propagation–detection interface is preserved. The two fibers cannot be quotient-identified merely because they generate a similar low-energy mass operator.

Proposition 11.5 (No fiber collapse by finite phenomenological agreement). *Agreement on a finite set of oscillation probabilities does not identify two completion fibers under Q_{phys} unless the target declares those probabilities to be the complete invariant bundle.*

Proof. The target retains field/operator scope, boundary behavior, environment coupling, anomaly and symmetry data, continuation range, and later observable interfaces. Two completions can agree on present oscillation records while differing in lepton-number violation, heavy states, nonunitarity, defect action, cosmology, or rare processes. Such differences are not skin unless shown irrelevant to the declared target. Finite fit agreement therefore gives empirical equivalence on a projection, not full target identity. $\square$

11.4 Beyond the benchmark fibers

The four imported fibers are benchmarks, not a complete candidate universe. A subsequent systematic classification may contain radiative completions, additional singlets, vector-like leptons, nonstandard interactions, open-system channels, composite or solitonic carriers, topological or defect-localized sectors, Lorentz- or CPT-modified propagation, extra dimensions, or other typed classes. The present theorem neither endorses nor excludes them.

Generalized and non-invertible symmetry data are deliberately absent from the necessity proof. They may become relevant to a complete physical-interface analysis, but only if their relevance is derived independently from the target rather than assumed through a preferred completion language.

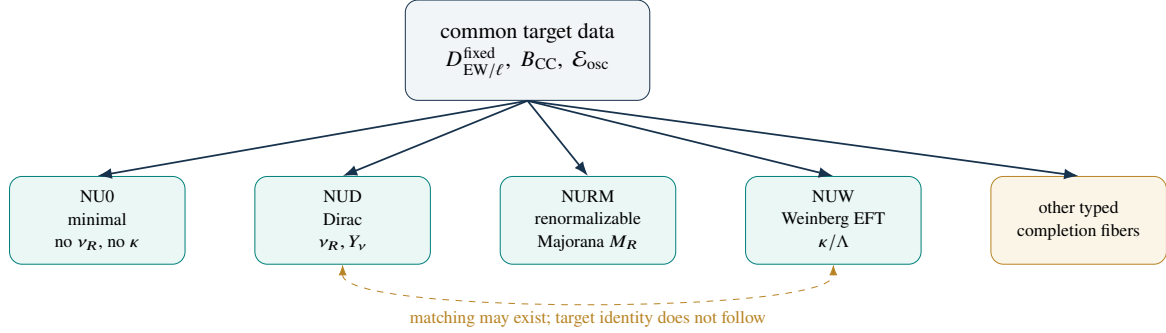


Figure 9: Common empirical and operational base with target-distinct completion fibers.

Open-family boundary

The family $\mathfrak{T}_\nu$ carries no probability or preference over completion fibers. It is only a typed comparison structure that keeps the completion coordinate open while preserving target identity within each candidate.

12 Robustness, scope conditions, and failure modes

12.1 Same-target challenge standard

A substantive counterexample must preserve the target it claims to defeat. In the present setting this means preserving the charged-current boundary, the empirical evidence of section 2, the stated phenomenological response assumptions, the one-completion condition, and the lawful comparison class, unless the counterexample explicitly challenges one of those target components. A theory with different source/detection interactions or additional target-individuating structure can be a legitimate alternative completion, but it is not automatically a same-target counterexample to the necessity theorem.

The principal challenge classes and their corresponding burdens are summarized below.

Table 8: Principal robustness tests.

Challenge	Required burden	Disposition
The target already assumes the standard mass–mixing solution.	Exhibit a mass matrix, PMNS entry, or fitted microscopic mass coordinate in the target definitions or role definitions.	Not present: the target records and transition kernels are operational.

Challenge	Required burden	Disposition
One nonlinear microscopic parameter can generate both visible response classes.	Exhibit such a completion.	Allowed; r_Δ constrains operational response multiplicity, not microscopic parameter count.
Source or detector new physics can mimic flavor change with aligned propagation.	Preserve the fixed charged-current boundary while changing no preparation or detection map.	If the maps change, the target changes; otherwise the channel theorem applies to the complete interface.
Decoherence or nonunitarity evades the commutator argument.	Give a CPTP channel that preserves every charged-current boundary projector while producing certified off-diagonal transition.	Excluded by theorem 12.1.
Wave-packet or finite-resolution effects invalidate ideal boundary projectors.	Show that calibrated source/detector impurity reproduces the admitted cross-boundary signal within an aligned contamination bound.	The exact theorem applies to the ideal quotient record; the robust form is conditional on the experimental contamination bounds.
A basis choice creates or removes the mismatch.	Change $[P_\alpha, U]$ or the transition kernels by simultaneous lawful redescription.	Excluded by theorems 4.7 and 8.7.
Internal spectral richness already occupies r_Δ .	Show that the hidden structure descends to at least two inequivalent boundary-visible response classes.	Excluded by theorem 8.3.
Nonalignment and response multiplicity are the same role.	Prove $r_{\text{b}\Delta} \Rightarrow r_\Delta$.	Refuted by the single-mode nonaligned model theorem 8.2.
The generic realized-physical-interior mass role already discharges r_Δ .	Supply a target-specific response witness in D_{SM}^{v0} .	Excluded by theorem 9.4.
Standard Model loops or ordinary matter effects generate the missing occupant.	Exhibit an admitted same-fiber operator and propagation kernel with cross-boundary support transfer.	No such structure exists in the declared renormalizable comparison fiber; added operators define a richer fiber.
The visible response multiplicity could reduce to one under a common-boundary reanalysis.	Provide a source-supported reanalysis, including its nuisance treatment, that identifies the two reactor responses as one lawful mode class.	Genuine reopening condition for r_Δ .

Challenge	Required burden	Disposition
The residual role complex implies a new particle.	Supply candidate completeness, joint compatibility, and unique role occupancy from the present theorem alone.	Not implied; those premises are absent.

12.2 Channel-level closure of the nonalignment result

The most direct mathematical challenge is to replace unitary propagation with a general quantum channel while preserving all charged-current boundary projectors. The following result closes that route.

Theorem 12.1 (Channel nonalignment theorem). *Let $\mathcal{E}$ be a completely positive trace-preserving map on $\mathcal{T}_1(\mathcal{H}_\nu)$. Let $\{P_\alpha\}$ be mutually orthogonal boundary projections and let $\rho_\alpha = P_\alpha \rho_\alpha P_\alpha$. If*

$$\mathcal{E}^\dagger(P_\beta) = P_\beta \quad \text{for every } \beta,$$

then

$$\text{Tr}(P_\beta \mathcal{E}(\rho_\alpha)) = 0 \quad (\beta \neq \alpha).$$

Consequently any effective channel realizing a certified cross-flavor transition must fail pointwise boundary-projector invariance or alter the boundary preparation/detection target.

Proof. For $\beta \neq \alpha$,

$$\text{Tr}(P_\beta \mathcal{E}(\rho_\alpha)) = \text{Tr}(\mathcal{E}^\dagger(P_\beta) \rho_\alpha) = \text{Tr}(P_\beta \rho_\alpha) = 0$$

by orthogonality and ideal support. The contrapositive gives the conclusion. $\square$

12.3 Explicit conditional assumptions

Three limitations are part of the theorem statement rather than hidden premises.

1. **Empirical response multiplicity.** The response theorem relies on the primary experimental analyses and their declared phenomenological response models to establish at least two nuisance-separated reactor phase-frequency classes. The paper does not independently reanalyze event-level data or claim a model-independent spectral decomposition.
2. **Effective-channel adequacy.** The theorem assumes that, after the stated source and detector corrections, the relevant experiment admits a well-defined source–propagation–detection channel representation. Standard QFT treatments support such reductions in the regimes considered [1, 2, 5, 8]; a theory denying the reduction would require a distinct target formulation.
3. **Comparison-fiber completeness.** The nonoccupancy theorem is complete only relative to the minimal renormalizable Standard Model fiber. Hidden sectors, quantum-gravity effects, or higher-dimensional operators are not claimed absent from nature; they are outside that comparison fiber.

Theorem 12.2 (Disposition of same-target challenges). *Every substantive challenge to the residual theorem has one of four logically distinct outcomes: it is rejected by a local theorem, absorbed as a lawful generalization that leaves the endpoint unchanged, shown to alter the target or comparison fiber, or retained as an explicit empirical/assumption limitation. No same-target challenge is resolved by redefining a failed candidate as the original target.*

Proof. A challenge either preserves the target and contests a theorem consequence, preserves the theorem while changing only presentation, changes target-individuating data, or disputes an input not proved locally. These are respectively theorem refutation, lawful redescription/generalization, target change, and explicit

conditionality. The alternatives exhaust the ways a challenge can relate to the fixed target and the proved role complex. $\square$

12.4 Scope-closure condition

Theorem 12.3 (Scope-closure condition). *The residual role complex of theorem 10.3 can be used as a premise for subsequent interface-completeness analysis only while:*

- (a) *the empirical propositions of section 2 remain admitted;*
- (b) *the channel-native nonalignment and response-multiplicity deletion theorems remain valid;*
- (c) *the strict relation $r_{\Delta} \succ r_{\bowtie}$ remains non-equivalent;*
- (d) *the known-substrate occupancy analysis remains complete at the declared scope; and*
- (e) *no faithful same-target counterexample is supplied.*

If any condition fails, the target, role complex, or theorem strength must be reconsidered before the residual conclusion is used downstream.

Proof. Failure of (a) reopens the target; failure of (b)–(c) reopens the role theorem; failure of (d) reopens the residual set difference; and failure of (e) directly defeats same-target necessity. The conditions therefore state exactly what must remain stable for the residual role conclusion to retain its stated scope. $\square$

13 Implications for completion and interface analysis

13.1 Data fixed by the present theorem

The residual theorem does not choose a physical completion. It fixes the information that any subsequent completion analysis must preserve. Let

$$H = (H01, \dots, H10)$$

denote the following collection.

ID	Content	Meaning
H01	Target fingerprint	$\mathfrak{f}(T_v^{\text{osc}})$, fixed anchors, open coordinates, and target-change rules.
H02	Empirical response data	The operational flavor records and finite visible-mode definition, including the stated environment/calibration assumptions.
H03	Target-specific roles	$r_{\bowtie}$ and r_{Δ} with carrier-neutral definitions.
H04	Role-order relation	$r_{\Delta} \succ r_{\bowtie}$; no inference of distinct occupants.
H05	Lawful quotient	Q_{phys} , simultaneous-redescription rules, and anti-skin invariants.
H06	Deletion witnesses	Boundary-preserving no-transfer and visible-multiplicity under-realization tests.
H07	Known occupancy	Upstream roles occupied; target-specific roles unoccupied in $D_{\text{SM}}^{\nu 0}$.
H08	Open completion family	Common base and benchmark fibers NU0, NUD, NURM, NUW, with no automatic role transfer between them.
H09	Prohibited assumptions	No mass, PMNS, ordering, particle, or generalized-symmetry assumption unless independently required by the target.
H10	Reopening conditions	Empirical response multiplicity, effective-channel adequacy, comparison-fiber boundary, and same-target counterexamples.

Table 9: Information fixed by the present role-necessity theorem.

13.2 The next mathematical problem

Any completion X capable of realizing the residual role complex must eventually be characterized by a complete physical-interface invariant of the schematic form

$$\Pi_{\nu}^{\text{phys}}(X) = (\Pi_{\text{boundary}}, \Pi_{\text{propagation}}, \Pi_{\text{response}}, \Pi_{\text{coherence}}, \Pi_{\text{environment}}, \Pi_{\text{symmetry}}, \Pi_{\text{anomaly}}, \Pi_{\text{global/defect}}, \Pi_{\text{observable}}, \dots).$$

The ellipsis is not permission to add arbitrary structure: each coordinate must have an independent physical definition, a lawful equivalence, and a variation or deletion test establishing its relevance to the target.

Theorem 13.1 (Sufficiency of the fixed role data for interface analysis). *The data in $\mathbf{H}$ are sufficient to formulate an occupant-neutral interface-completeness problem for the residual role complex. Every load-bearing result of this paper is represented in $\mathbf{H}$, and no entry asserts carrier identity or realization.*

Proof. The target and empirical basis are H01–H02. The work to be reproduced and its internal relation are H03–H04. Identity and deletion tests are H05–H06. The known comparison fiber and the open family are H07–H08. H09 prevents completion-specific backflow, and H10 preserves the exact theorem boundary. Together these data determine the interface problem while leaving field content, operator class, dimensionality, locality type, and all numerical parameters open. $\square$

Corollary 13.2 (No completion-specific coordinate may be added by convenience). *A generalized-symmetry coordinate, mass parameter, or preferred completion may enter a subsequent interface analysis only if it is independently shown necessary for reproducing the fixed target or is declared as part of a different target.*

13.3 Interface-completeness questions

A complete successor analysis must answer, without assuming the standard mass–mixing solution:

1. What physical data distinguish lawful occupants of $\mathfrak{R}_{\nu}^{\text{osc}}$?
2. Which data are invariant under Q_{phys} , and which are presentation skin?
3. How are source, propagation, detector, matter/environment, coherence, global-form, and defect data coupled?
4. Which interface coordinates are required by the target rather than by a chosen candidate language?
5. What would constitute an occupancy-positive signature outside the generated interface range?

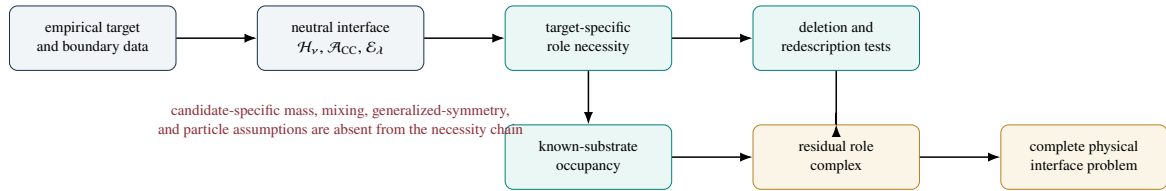


Figure 10: Dependency structure from empirical target through role necessity and residual nonoccupancy to the subsequent interface-completeness problem.

Subsequent problem

Construct, without naming a preferred carrier, the complete lawful physical-interface invariant whose realization is necessary and sufficient for a completion to reproduce the fixed target and occupy the nested role complex $r_{\Delta} \succ r_{\nu\Delta}$.

14 Conclusion

The argument begins from a realized physical interface rather than from a proposed neutrino model. The fixed target specifies operational charged-current source and detector channels, established flavor transformation, path/energy-dependent modulation, and a common reactor electron-antineutrino survival subinterface whose stated phenomenological response analyses contain at least two nuisance-separated mode classes. The microscopic completion remains open. A framework-neutral quantum-channel description, with unitary propagation only as a representative special case, then makes the role question precise.

The first local result is elementary but load-bearing: a propagation channel that preserves every charged-current boundary face cannot transport an ideally prepared state into a different boundary sector. Flavor transformation therefore requires interaction–propagation nonalignment. The second result is physically distinct: under the stated calibration and environment assumptions, the common reactor survival interface has visible response multiplicity

$$\rho_{\text{vis}}(C_\nu) \geq 2.$$

Any realization of the fixed target must reproduce that multiplicity. This is an operational response requirement, not a count of data sets, masses, gaps, fields, particles, couplings, or microscopic parameters.

The relation between the two coordinates is strict. Visible response multiplicity of at least two entails nonalignment because a boundary-face-preserving channel makes the ideal reactor survival response protocol-trivial. The converse fails: the explicit two-state nonaligned model has one nonconstant visible modulation class and therefore $\rho_{\text{vis}} = 1$. The target-specific role structure is thus

$$r_\Delta \succ r_{\triangleright\triangleleft},$$

a strict nested relation between two non-equivalent pieces of physical work.

Only after that role complex has been derived is the minimal Standard Model comparison fiber tested for occupancy. The realized physical interior and Standard Model structural results supply the general field, particle, gauge, transport, vacuum, record, and charged-current boundary architecture. The minimal renormalizable massless-neutrino fiber supplies no standing PMNS object, no boundary-sector transfer, and no nontrivial boundary-visible response class. The residual theorem is therefore

$$R_{\text{res}}(C_\nu, T_\nu^{\text{osc}}; D_{\text{SM}}^{\nu 0}) = (\{r_{\triangleright\triangleleft}, r_\Delta\}, r_\Delta \succ r_{\triangleright\triangleleft}) \neq \emptyset.$$

Theorem 14.1 (Synthesis theorem). *Conditional on the primary experimental evidence, the phenomenological reactor-response assumptions, the effective source–propagation–detection comparison class, and the imported fixed-domain discipline stated in this paper:*

- (i) *the realized neutrino oscillation target requires channel-native interaction–propagation nonalignment;*
- (ii) *it requires one completion to carry boundary-visible propagation-response multiplicity at least two;*
- (iii) *the response-multiplicity burden strictly entails, but is not equivalent to, the nonalignment burden;*
- (iv) *both coordinates are invariant under lawful simultaneous redescription and cannot be generated by weak-basis or PMNS-chart choice;*
- (v) *the minimal renormalizable massless-neutrino Standard Model occupies neither target-specific coordinate; and*
- (vi) *the resulting residual role complex is nonempty while its physical occupant remains undetermined.*

14.1 Falsification and reopening conditions

The result can be defeated or narrowed by any of the following:

1. a corrected primary analysis invalidating the admitted flavor-transition evidence or the two-class reactor response statement;
2. a same-boundary effective channel that preserves every boundary face while nevertheless producing the certified cross-boundary transition beyond the declared contamination envelope;
3. a lawful common-boundary calibration/environment equivalence or carrier-neutral response reconstruction that reduces the visible reactor response multiplicity to one;
4. a demonstration that the single-mode nonaligned countermodel lies outside the stated comparison class, thereby reopening strict role non-equivalence; or
5. an explicit target-specific occupant of the role complex inside the minimal renormalizable massless-neutrino Standard Model fiber.

A candidate in a richer completion fiber does not falsify the theorem; it is a possible realization of the residual role content.

14.2 Nonclaims

The theorem does not establish that nature contains physics beyond the Standard Model in every possible sense. It establishes only that the sharply defined minimal massless-neutrino Standard Model fiber does not realize an empirically instantiated target-specific role complex. It does not infer that the occupant is local propagating matter; does not determine its spin, chirality, representation, multiplicity, mass character, or scale; does not select Dirac or Majorana neutrinos; does not derive the PMNS matrix, mixing angles, CP phases, mass ordering, or absolute mass; and does not require generalized or non-invertible symmetry.

The visible-response result is conditional on the collaborations' phenomenological response models and calibrated frequency extractions. The paper does not re-fit raw event data or claim that $\rho_{\text{vis}} \geq 2$ is a model-independent Fourier fact. A future common-boundary analysis that invalidated the stated response assumptions or represented both reactor responses by one lawful mode class would reopen r_{Δ} .

The paper also does not invoke the sufficiency direction of Role-Occupancy Closure. That theorem becomes relevant only after candidate completeness, quotient completeness, eligibility completeness, joint compatibility, live-residue completeness, and failure completeness have been independently established.

Theorem boundary

The result closes a physical role-necessity problem, not a particle-prediction problem. The realized neutrino oscillation interface requires target-specific physical work absent from the minimal massless-neutrino Standard Model comparison fiber. The identity of the physical structure that performs that work remains an open completion problem.

A Formal target specification

This appendix collects the target data used throughout the paper in one place. It is a mathematical summary of the physical scope, not an additional premise beyond the definitions and empirical assumptions already stated in the main text.

A.1 Target identity

Field	Fixed content
Target	Realized neutrino source–propagation–detection oscillation interface
Comparison fiber	Minimal renormalizable Standard Model neutrino fiber $D_{\text{SM}}^{\nu^0}$ for the known-occupancy comparison
Background	Four-dimensional local relativistic electroweak/lepton substrate; experiment-specific matter/environment profiles stated per record
Boundary	Known charged-current production and detection classes tagged by associated charged leptons
Physical quotient	Simultaneous lawful redescription preserving complete transition kernels, boundary semantics, phenomenological reactor-response assumptions, lawful response-mode classes, environment/calibration data, and the target fingerprint
Completion policy	Open: no mass operator, field extension, PMNS chart, Dirac/Majorana choice, or generalized-symmetry completion is fixed
Empirical policy	Primary-source qualitative records; global-fit numerical coordinates are not structural premises
Target-change rule	A change of field content, boundary instrument, operator order, response equivalence, or empirical record defines a distinct target or target variant

Table 10: Fixed target data.

A.2 Target tuple

The target is

$$T_{\nu}^{\text{osc}} = \left(D_{\text{EW}/\ell}^{\text{fixed}}, \mathcal{B}_{\text{CC}}, \mathcal{P}_{\text{rec}}, \mathcal{E}_{\text{osc}}, \mathcal{Q}_{\text{phys}}, C_{\text{open}} \right).$$

Each coordinate is typed as follows.

Table 11: Target-coordinate dictionary.

Coordinate	Role	Fixed content
$D_{\text{EW}/\ell}^{\text{fixed}}$	Known substrate	Electroweak gauge/lepton source and detector architecture, charged-lepton channel individuation, local relativistic quantum description, and the minimal comparison fiber used only for occupancy analysis.
$\mathcal{B}_{\text{CC}}$	Operational boundary	Source and detector maps whose active flavor tags are defined by associated charged leptons; backgrounds and misidentification are included in the empirical response descriptions.
$\mathcal{P}_{\text{rec}}$	Propagation comparison class	Effective boundary-to-boundary quantum channel family with a coherent unitary representative in the algebraic examples; environment/path data are explicit protocol coordinates.

Coordinate	Role	Fixed content
$\mathcal{E}_{\text{osc}}$	Empirical evidence	Flavor transformation, path/energy modulation, at least two nuisance-separated reactor response-mode classes under the stated phenomenological response assumptions, and cross-protocol stability.
$\mathcal{Q}_{\text{phys}}$	Lawful equivalence	Simultaneous redescription preserving the complete target fingerprint, phenomenological response-model class, and lawful mode-equivalence relation.
$\mathcal{C}_{\text{open}}$	Completion coordinate	Field content, operator load, mass character, state-space dimension, global/defect data, and numerical parameters remain open.

A.3 Fixed and open coordinates

Coordinate	Status	Effect of changing it
Charged-current boundary semantics	Fixed	A new source/detection interaction or flavor tag defines a target variant.
Primary flavor-transition evidence	Fixed empirical input	Removal or reversal reopens r_{pd} .
Propagation-dependent modulation	Fixed empirical input	If no lawful modulation remains, the propagation-response target must be weakened.
At least two lawful reactor response-mode classes	Fixed conditionally on the stated response analyses	A lawful common-boundary reconstruction showing one mode class reopens r_{Δ} .
Environment/calibration data	Fixed per record	Updated calibration requires re-evaluation of the corresponding target record.
Neutrino mass operator	Open	May enter only as a completion/fiber coordinate.
PMNS coordinates	Open/representative	Not a target anchor.
Dirac/Majorana status	Open	Distinct completion fibers.
Additional fields/sectors	Open	Each addition defines an explicit candidate or target variant.
Global, higher-form, non-invertible data	Open	May become relevant only if independently required by a complete interface analysis.
Numerical mass, mixing, CP, ordering, scale	Open	Not inferred here.

Table 12: Fixed and open target coordinates.

A.4 Empirical propositions

Table 13: Empirical propositions used in the target.

ID	Proposition	Minimal content	Status
ER-0	Operational channel individuation	Production/detection flavor labels are fixed by charged-current association with e , μ , or τ at the boundary.	Known substrate
ER-1	Nontrivial flavor transformation	At least one source-defined flavor class yields an admitted detector record outside the same boundary class, or an equivalent appearance record establishes cross-channel transfer.	Empirical
ER-2	Propagation-dependent modulation	The admitted transition/survival response depends nontrivially on path and/or energy and is not exhausted by a constant source-normalization shift.	Empirical
ER-3	Two lawful reactor response-mode classes	Under the collaborations' stated response models and nuisance treatment, the KamLAND and Daya Bay phase-frequency intervals define at least two inequivalent classes in $\mathfrak{M}_{\text{vis}}(C_\nu)$, so $\rho_{\text{vis}}(C_\nu) \geq 2$.	Empirical/model conditional
ER-4	Cross-protocol mode stability	The distinction survives common-unit conversion and source-authorized calibration/environment transports; otherwise ER-3 reopens.	Empirical/model conditional
ER-5	One-completion condition	One physical completion must reproduce the complete target rather than assigning unrelated theories to different records.	Target condition

A.5 Target fingerprint

The target fingerprint is

$$\mathfrak{f}(T_\nu^{\text{osc}}) = (D_{\text{EW}/\ell}^{\text{fixed}}, \mathcal{B}_{\text{CC}}, \{\text{ER}0, \dots, \text{ER}5\}, \mathfrak{P}_e^{\text{phen}}, \mathfrak{M}_{\text{vis}}(C_\nu), \approx_{\text{mode}}, \mathcal{P}_{\text{env/cal}}, \mathcal{Q}_{\text{phys}}, C_{\text{open}}).$$

A same-target transformation is lawful only if it preserves this invariant content up to the equivalence defined in the main text.

A.6 Target changes

A new target analysis is required when a proposal alters any of:

1. source or detector interaction type;
2. empirical record standing;
3. response equivalence or mode-class identity;
4. field/operator content of the comparison fiber;
5. environment or background type outside the declared protocol family;

6. lawful quotient or record-use semantics;
7. the one-completion condition.

No such change inherits the residual-role conclusion automatically.

B Primary empirical evidence

The experimental publications are used to instantiate the operational target, not to select a microscopic completion. The summaries below state only the qualitative propositions needed in the theorem and the limits of what is imported from each source.

B.1 Atmospheric neutrinos: Super-Kamiokande

Atmospheric evidence

Source. Super-Kamiokande’s atmospheric-neutrino zenith-angle analysis reported a flavor-dependent deficit inconsistent with a no-oscillation description and consistent with neutrino oscillation [29]. The later L/E analysis identified an oscillatory signature rather than a monotone attenuation-only pattern [28].

Use here. A muon-associated source/detection class exhibits path/energy-dependent survival behavior in the atmospheric regime, supporting nontrivial flavor transformation and propagation-dependent modulation.

Not imported. No best-fit Δm^2 , mixing amplitude, two-flavor exactness, mass ordering, or PMNS coordinate is used as a structural premise.

B.2 Solar flavor transformation: SNO

Solar evidence

Source. SNO compared charged-current sensitivity to electron neutrinos with neutral-current sensitivity to all active flavors and reported direct evidence for solar neutrino flavor transformation [25].

Use here. The source-defined solar electron-neutrino class does not remain wholly in the electron boundary class, independently supporting cross-boundary flavor transformation.

Not imported. The standard MSW interpretation, a solar mixing angle, and a numerical splitting are not used to define either target-specific role.

B.3 Long-baseline reactor response: KamLAND

KamLAND evidence

Source. KamLAND observed reactor electron-antineutrino disappearance with spectral distortion and later reported precision oscillation measurements over long baselines [6, 7].

Use here. The reactor electron-antineutrino survival response contains a source-supported slow phase-frequency interval. Together with Daya Bay, this supplies one of the two nuisance-separated response-mode classes used in the visible-multiplicity result.

Not imported. The reported Δm_{21}^2 is used only as the collaboration’s conventional coordinate for the measured phase-frequency scale; it is not a microscopic role definition or carrier count.

B.4 Short-baseline reactor response: Daya Bay

Daya Bay evidence

Source. Daya Bay observed short-baseline electron-antineutrino disappearance and measured both oscillation amplitude and frequency through spectral comparison [3, 4].

Use here. The same charged-current electron-antineutrino boundary exhibits a source-supported faster phase-frequency interval under Daya Bay’s stated phenomenological response model and nuisance/calibration treatment. Together with the KamLAND interval, this supplies the second lawful mode class used in the response-multiplicity theorem.

Not imported. No numerical θ_{13} , three-neutrino state count, or microscopic interpretation of the reported frequency coordinate is used as a premise.

B.5 Appearance cross-checks

Tau-appearance evidence

Sources. Super-Kamiokande reported evidence for atmospheric tau-neutrino appearance, and OPERA reported tau-neutrino appearance in a muon-neutrino beam [24, 27].

Use here. These results corroborate the operational reading of flavor transformation. The nonalignment theorem does not depend uniquely on either experiment.

Not imported. No assumption is made that the only relevant channel is $\nu_\mu \rightarrow \nu_\tau$ or that a particular unitary three-state model is complete.

B.6 Synthesis of the empirical input

Proposition B.1 (Empirical evidence synthesis). *The atmospheric, solar, long-baseline reactor, and short-baseline reactor results jointly support the target propositions of nontrivial flavor transformation, propagation-dependent modulation, and at least two separated reactor response-mode classes without requiring a global-fit parameter table as an input to the necessity theorem.*

Proof. Super-Kamiokande and SNO establish flavor-dependent transition or disappearance in physically distinct source and path regimes. KamLAND and Daya Bay share the same reactor electron-antineutrino boundary semantics and resolve phase-frequency intervals separated by more than an order of magnitude under the collaborations’ calibrated analyses. Under the explicit response-model assumptions stated in section 2, those intervals remain disjoint after the admitted nuisance treatment and therefore define two classes in $\mathfrak{M}_{\text{vis}}(C_\nu)$. Solar and atmospheric experiments independently support the larger flavor-transformation target. The collaborations conventionally report mass-squared parameters, and Daya Bay states a three-neutrino response framework for its frequency extraction; these are retained as empirical-analysis coordinates, not identified with microscopic carrier cardinality. $\square$

Evidence boundary

This appendix is not a new statistical combination and does not supersede the experimental collaborations’ analyses. Exact confidence levels, systematics, and current global-fit refinements remain in the primary literature and are not rederived here.

C Proof details and controlled countermodels

C.1 General invariant-block channel theorem

The main channel theorem fixes every boundary projector pointwise. A slightly more general formulation uses invariant blocks.

Theorem C.1 (Invariant-block no-transfer theorem). *Let $\{P_\alpha\}_{\alpha \in A}$ be mutually orthogonal projections and $\mathcal{E}$ a CPTP map. Suppose the face*

$$\mathfrak{S}_\alpha = \{\rho \geq 0 : \text{Tr } \rho = 1, \rho = P_\alpha \rho P_\alpha\}$$

is invariant under $\mathcal{E}$ for each α . Then every off-block transition probability vanishes:

$$\text{Tr}(P_\beta \mathcal{E}(\rho_\alpha)) = 0 \quad (\rho_\alpha \in \mathfrak{S}_\alpha, \beta \neq \alpha).$$

Proof. Invariance gives $\mathcal{E}(\rho_\alpha) = P_\alpha \mathcal{E}(\rho_\alpha) P_\alpha$. Orthogonality then yields

$$P_\beta \mathcal{E}(\rho_\alpha) = P_\beta P_\alpha \mathcal{E}(\rho_\alpha) P_\alpha = 0.$$

Taking the trace proves the claim. $\square$

Pointwise Heisenberg fixation $\mathcal{E}^\dagger(P_\alpha) = P_\alpha$ is a sufficient condition for the corresponding sector probabilities to be conserved. The role theorem needs only failure of invariant-block preservation somewhere in the complete interface.

C.2 Robust aligned-boundary estimate

Let $P = P_\alpha$, $Q = P_\beta$ with $PQ = 0$. Suppose $\text{Tr}(P\rho) \geq 1 - \epsilon$ and the ideal aligned propagation satisfies $[P, U] = [Q, U] = 0$. Let E_β be an actual detector effect with $\|E_\beta - Q\|_\infty \leq \eta$.

The gentle measurement estimate gives

$$\|\rho - P\rho P\|_1 \leq 2\sqrt{\epsilon}.$$

Then

$$\begin{aligned} \text{Tr}(E_\beta U \rho U^\dagger) &\leq |\text{Tr}[QU(\rho - P\rho P)U^\dagger]| + |\text{Tr}[(E_\beta - Q)U\rho U^\dagger]| \\ &\quad + \text{Tr}(QU P \rho P U^\dagger). \end{aligned}$$

The last term vanishes by alignment and orthogonality. Hölder's inequality bounds the first term by $2\sqrt{\epsilon}$ and the second by η . Therefore

$$\text{Tr}(E_\beta U \rho U^\dagger) \leq 2\sqrt{\epsilon} + \eta.$$

For a general channel preserving the P -face, the same proof applies with $U\rho U^\dagger$ replaced by $\mathcal{E}(\rho)$ and trace-norm contractivity used on the leakage term.

This is a structural robustness bound, not an experimental analysis. Each experiment must supply its own calibrated ϵ , η , background model, and record standing.

C.3 Path-ordered diagonal matter evolution

Suppose

$$H(x) = \sum_\alpha h_\alpha(x) P_\alpha$$

with fixed mutually orthogonal projectors. Since $[H(x), P_\alpha] = 0$ for every x , the path-ordered exponential

$$U(L) = \mathcal{P} \exp\left(-i \int_0^L H(x) dx\right)$$

commutes with every P_α . Indeed each product in the Dyson expansion commutes with every projector. Hence spatially varying diagonal matter potentials do not transport support between flavor sectors. Off-diagonal matter interactions would alter the candidate fiber and can occupy nonalignment in their own

right.

C.4 Countermodel CM-B in full

For the two-state model of theorem 8.2,

$$U_t = \begin{pmatrix} \cos \omega t & -i \sin \omega t \\ -i \sin \omega t & \cos \omega t \end{pmatrix}.$$

With $\rho_0 = P_0$,

$$K_{00} = \cos^2 \omega t, \quad K_{10} = \sin^2 \omega t.$$

There is nonalignment because

$$[P_0, U_t] = \begin{pmatrix} 0 & -i \sin \omega t \\ i \sin \omega t & 0 \end{pmatrix}$$

is nonzero for generic t . Yet the target record of this countermodel has only one protocol family and one kernel shape up to the declared equivalence. This model is sufficient to prove $r_{\bowtie} \not\Rightarrow r_{\Delta}$.

C.5 Why an aligned two-response countermodel is impossible here

If every charged-current boundary face is invariant under the effective channel, then every ideal source state remains in its boundary sector. The survival probabilities are protocol-trivial and transition probabilities vanish, apart from fixed boundary imperfections. Hence the boundary-visible mode set is empty, $\mathfrak{M}_{\text{vis}} = \emptyset$, and $\rho_{\text{vis}} = 0$. An aligned system may have arbitrarily rich hidden spectra, but internal modes that do not descend to a nontrivial protocol-resolved response class contribute nothing to $\mathfrak{M}_{\text{vis}}$. This proves the forward implication $r_{\Delta} \Rightarrow r_{\bowtie}$ without identifying response multiplicity with microscopic spectral multiplicity.

C.6 Lawful conjugation and commutants

Let V be unitary. Then

$$[VPV^\dagger, VUV^\dagger] = V[P, U]V^\dagger.$$

Thus commutation is invariant. Similarly,

$$\text{alg}\{VP_\alpha V^\dagger\} = V\mathcal{A}_{\text{CC}}V^\dagger, \quad \text{vN}\{VU_\lambda V^\dagger\} = V\mathcal{A}_{\text{prop}}V^\dagger,$$

and

$$V\mathcal{A}_{\text{CC}}V^\dagger \subseteq (V\mathcal{A}_{\text{prop}}V^\dagger)' \iff \mathcal{A}_{\text{CC}} \subseteq \mathcal{A}'_{\text{prop}}.$$

These equations are the algebraic form of the anti-skin theorem.

C.7 QFT comparator boundary

In a QFT treatment, production and detection are localized interaction regions and the oscillation amplitude is obtained from the internal neutrino propagator integrated against external wave packets. Under the standard localization and coherence conditions, the result factors into process-dependent boundary amplitudes and a propagation-dependent interference kernel [1, 2, 5, 8]. The present analysis uses the effective kernel after that reduction. The no-transfer theorem is therefore conditional on the existence of a well-defined boundary-sector channel, not on plane waves or exact flavor eigenstates.

A QFT completion can evade the particular finite-dimensional unitary representative in several ways: nonorthogonal wave packets, unstable modes, unobserved states, environment entanglement, or source/detector interactions. It cannot evade the role statement while preserving the same effective target: either the complete channel transports probability between boundary classes, which is nonalignment, or the operational boundary/record is changed.

D Standard mass–mixing comparator

This appendix shows how the conventional three-neutrino description realizes the target-specific role complex. It is a comparator only: no equation in this appendix is used to establish the role definitions or the residual theorem.

D.1 Vacuum representative

In the conventional unitary description, charged-current flavor states and propagation mass states are related by

$$|\nu_\alpha\rangle = \sum_{i=1}^3 U_{\alpha i}^* |\nu_i\rangle,$$

where U is the PMNS matrix in a chosen phase convention [10]. Propagation over baseline L at energy E gives relative phases

$$|\nu_i(L)\rangle \sim \exp\left[-i\frac{m_i^2 L}{2E}\right] |\nu_i(0)\rangle.$$

The amplitude is

$$\mathcal{A}_{\alpha\rightarrow\beta}(L, E) = \sum_i U_{\beta i} \exp\left[-i\frac{m_i^2 L}{2E}\right] U_{\alpha i}^*,$$

and the probability can be written

$$\begin{aligned} P_{\alpha\rightarrow\beta} &= \delta_{\alpha\beta} - 4 \sum_{i>j} \Re\left(U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j}\right) \sin^2 \Delta_{ij} \\ &\quad + 2 \sum_{i>j} \Im\left(U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j}\right) \sin 2\Delta_{ij}, \end{aligned}$$

with

$$\Delta_{ij} = \frac{\Delta m_{ij}^2 L}{4E}.$$

In this representative, a nontrivial mismatch between the charged-current flavor projectors and the mass-basis propagation generator occupies r_{ps} . The independent relative phase structure can generate at least two boundary-visible response classes and thereby occupy r_{Δ} when $\rho_{\text{vis}} \geq 2$. The same matrix-and-spectrum structure can therefore perform both roles; the role theorem does not require two distinct carriers.

D.2 Matter representative

Coherent forward scattering contributes the Wolfenstein matter potential [31]; the solar resonance behavior was developed by Mikheyev and Smirnov [23]. In flavor coordinates the effective Hamiltonian is schematically

$$H_f(x) = \frac{1}{2E} U \text{diag}(m_1^2, m_2^2, m_3^2) U^\dagger + \text{diag}(V_e(x), 0, 0) + V_{\text{NC}}(x) \mathbf{1}.$$

Matter changes the propagation response but does not remove the boundary/propagation role distinction. If the vacuum term is proportional to the identity and no off-diagonal interaction is added, the Hamiltonian is diagonal in the boundary basis and cannot produce flavor conversion. Flavor-changing matter interactions, if introduced, would themselves contribute to nonalignment and therefore belong to a richer completion fiber.

D.3 Representative and invariant content

Standard object	Type in the present analysis	Status
PMNS matrix entries	Representative coordinates	Basis/phase dependent individually; no numerical values derived.
Rephasing-invariant products	Physical witness in a fixed completion	May enter a complete interface characterization if independently required.
Mass eigenvalue labels/order	Representative plus physical ordering data	Labels are conventional; ordering and gaps require separate physical definition.
Relative propagation phases	Completion-level response structure	One conventional way to realize r_Δ .
Flavor/mass incidence	Completion-level mismatch	One conventional way to realize $r_{\triangleright\triangleleft}$.
Oscillation kernels	Physical quotient content	Directly comparable with the fixed empirical records.
Matter potential	Declared environment contribution	Part of the protocol; not a selector among completions.

Table 14: Conventional mass–mixing quantities translated into the role language used in this paper.

D.4 Why the comparator does not establish uniqueness

The conventional description is empirically successful, but that success alone does not establish that it is the unique completion of the target. Other completions could reproduce the presently admitted oscillation kernel while differing in unitarity, field content, lepton-number violation, heavy sectors, topological structure, or additional observables. Establishing uniqueness would require a complete candidate and compatibility analysis beyond the necessity theorem proved here.

E Theorem and source provenance

This appendix records the principal logical dependencies of the local theorem chain and the scope of the imported results. It is included to make the dependency structure explicit while keeping the theorem statements focused on their mathematical and physical content.

E.1 Local theorem structure

Table 15: Principal local results and dependencies.

Result	Role in the argument	Principal dependencies
Carrier-neutrality of role necessity	Prevents completion-specific structure from entering the target-role definition.	Fixed-domain discipline; Role-Occupancy Closure ordering.
Lawful-redescription invariance	Shows that kernels, boundary-face invariance, and visible response multiplicity descend to the physical quotient.	Neutral interface definitions.

Result	Role in the argument	Principal dependencies
Commuting-boundary no-transition result	Excludes cross-boundary transfer under aligned unitary propagation.	Boundary projectors, ideal preparation, unitary propagation.
Channel nonalignment theorem	Generalizes no-transfer to CPTP channels.	Channel duality and boundary-projector invariance.
Nonalignment deletion theorem	Shows that removing $r_{\bowtie}$ destroys the flavor-transition target.	Empirical flavor transformation; channel theorem.
Boundary-visible response-multiplicity theorem	Establishes the r_{Δ} burden under the stated reactor-response assumptions.	KamLAND/Daya Bay response intervals; $\mathfrak{M}_{\text{vis}}$ and ρ_{vis} .
Response-multiplicity deletion theorem	Separates r_{Δ} from mere record coverage by deleting candidate-side response structure while holding the target fixed.	Same-target response reduction; fixed calibration and analysis maps.
Strict nested role theorem	Proves $r_{\Delta} \Rightarrow r_{\bowtie}$ and $r_{\bowtie} \not\Rightarrow r_{\Delta}$.	Single-mode nonaligned countermodel; no-transfer theorem.
Anti-skin theorem	Excludes basis, phase, and chart changes as generators of the target-specific roles.	Lawful-redescription invariance.
Upstream-role discharge	Separates general physical roles from target-specific neutrino roles.	Realized Physical Interior; Standard Model structural results.
Minimal-fiber nonoccupancy	Shows that $D_{\text{SM}}^{\nu 0}$ occupies neither $r_{\bowtie}$ nor r_{Δ} .	Minimal neutrino fiber; no-PMNS guard; local propagation fingerprint.
Residual role-obligation theorem	Forms the nonempty residual role complex.	Target-role composition; upstream discharge; minimal-fiber nonoccupancy.
Open-family well-typing	Keeps target-inequivalent completion fibers distinct.	SYM target-fiber separation; fixed-domain transfer discipline.

E.2 Imported theorem uses

Table 16: Scope of the principal imported theorem families.

Source	Source locus	Use here	Not imported
Non-Degenerate Construction and the Kernel of Admissibility	Lemma 2.6; Thms. 2.34, 6.20	Fixed target, no repair, no lower generator, anti-weakening, standing conservation	Neutrino physics or empirical roles
AMetric Boundary	Thms. 5.26, 14.2	No hidden selector/parameter at the role boundary; export discipline	Numerical or physical completion
Unus Solus Possibilis Est	Thms. 6–7	Unique admissible interior once existence and target are independently fixed	Existence of a carrier
Realized Physical Interior	Thms. 13.1, 14.1, 16.1, 20.1, 26.1, 38.2	General field, particle, response, record, and quantum-continuation roles	Target-specific neutrino occupancy
Role-Occupancy Closure	Def. 3.5; Thms. 4.1, 6.1, 7.4	Role set before residue; candidate non-realization; later completeness boundary	Sufficiency or unique occupant
SR1 simultaneous collapse	Thms. 7.3, 8.2, 9.4, 10.1	Warning that independent constraints must share one witness structure	Standard Model-specific survivor identification
SYM-5	No-PMNS guard; neutrino-target distinctions	Basis/PMNS non-primitivity and fiber typing	Numerical mixing coordinates
SYM-7	Four-fiber separation theorem	NU0, NUD, NURM, NUW target inequivalence	Preference among fibers
SM-5	PMNS non-primitivity; oscillation-route status	Minimal-fiber comparison and boundary-reconstruction discipline	Observed PMNS as primitive input
SM-6/SYM-11	Declared-scope Standard Model and symmetry closure	Target-change discipline for added fields/operators	Claim that every physical extension is excluded
Generation-cardinality endpoint	Forced role-cardinality precedent	Methodological precedent for fixing role load before an open coordinate	Direct neutrino conclusion

F Glossary and symbol register

Table 17: Glossary and symbol register.

Term/symbol	Meaning in this paper
AASC	Admissibility And Standing Constraint, the fixed-domain necessity/constraint formalism used for target and role governance.

Term/symbol	Meaning in this paper
C_ν	Fixed target data for the realized neutrino oscillation interface.
T_ν^{osc}	Fixed source–propagation–detection target.
$\mathcal{H}_\nu$	Record-relevant propagation Hilbert space; no fundamental dimension assumed.
P_α	Ideal quotient-level charged-current boundary projector for flavor tag α .
ρ_α	Boundary preparation supported in the α sector in the ideal representative.
$\mathcal{E}_\lambda$	Effective quantum channel for protocol λ .
U_λ	Coherent unitary representative of $\mathcal{E}_\lambda$ where applicable.
$K_{\beta\alpha}(\lambda)$	Boundary-to-boundary transition kernel.
$\mathcal{A}_{\text{CC}}$	Operational algebra generated by active charged-current boundary projectors.
$\mathcal{A}_{\text{prop}}$	von Neumann algebra generated by record-relevant propagation operators in the unitary representative.
$\mathcal{A}'_{\text{prop}}$	Commutant of the propagation algebra.
$\mathcal{Q}_{\text{phys}}$	Lawful physical quotient preserving complete kernels, boundary semantics, environment data, response equivalence, and target fingerprint.
$\text{Resp}(C_\nu)$	Auxiliary quotient of admitted calibrated response records; not the definition of r_Δ .
$\mathfrak{M}_{\text{vis}}$	Set of nontrivial protocol-resolved phase-frequency mode records modulo lawful calibration/nuisance equivalence. A finite uncertainty interval is one mode object.
ρ_{vis}	Boundary-visible response multiplicity $ \mathfrak{M}_{\text{vis}} $; for finite mode sets it is equivalently $\dim \mathbb{C}[\mathfrak{M}_{\text{vis}}]$.
$r_{\triangleright\triangleleft}$	Interaction–propagation nonalignment role.
r_Δ	Boundary-visible propagation-response-multiplicity role, requiring $\rho_{\text{vis}} \geq 2$ for the fixed target.
$r_\Delta \succ r_{\triangleright\triangleleft}$	Strict role implication: visible response multiplicity at least two entails nonalignment, while single-mode nonaligned models exist.
R_{upstream}	Relevant upstream realized-physical-interior role families.
$R_{\text{occupied}}^{\text{known}}$	Roles occupied by the known minimal Standard Model comparison substrate.
R_{res}	Target roles not occupied by that substrate, retaining precedence and lawful-equivalence data.
$D_{\text{SM}}^{\nu 0} / \text{NU0}$	Minimal renormalizable Standard Model neutrino fiber: no ν_R , no Weinberg operator, massless active neutrinos.
NUD	Dirac-neutrino completion fiber with ν_R and Dirac Yukawa structure.
NURM	Renormalizable Majorana/seesaw-capable fiber with ν_R and M_R .
NUW	Low-energy Weinberg-operator EFT fiber.
Role	Target-relative physical work defined by invariant content, deletion consequence, and carrier-neutral witness type.

Term/symbol	Meaning in this paper
Occupant	A physical structure or realization that performs a target role under a complete, jointly compatible realization.
Visible response mode	A protocol-resolved phase-frequency record that survives lawful mode equivalence under the stated phenomenological response and nuisance/calibration assumptions.
Target change	Explicit alteration of target-individuating data followed by a fresh target analysis.
Skin	Representative variation that changes no standing-bearing target content.
Empirical input	Primary-source observation used to instantiate the target; not a selector among candidate completions.
One-completion condition	Requirement that one physical completion reproduce the complete target; a split collection requires explicit target retyping.
Carrier-neutral	Formulated without naming or privileging an eventual physical occupant.

References

- [1] E. K. Akhmedov and J. Kopp. “Neutrino Oscillations: Quantum Mechanics versus Quantum Field Theory”. In: *Journal of High Energy Physics* 04 (2010), p. 008. DOI: [10.1007/JHEP04\(2010\)008](https://doi.org/10.1007/JHEP04(2010)008). arXiv: [1001.4815](https://arxiv.org/abs/1001.4815) [hep-ph].
- [2] M. Beuthe. “Oscillations of Neutrinos and Mesons in Quantum Field Theory”. In: *Physics Reports* 375 (2003), pp. 105–218. DOI: [10.1016/S0370-1573\(02\)00538-0](https://doi.org/10.1016/S0370-1573(02)00538-0). arXiv: [hep-ph/0109119](https://arxiv.org/abs/hep-ph/0109119).
- [3] Daya Bay Collaboration, F. P. An, et al. “Observation of Electron-Antineutrino Disappearance at Daya Bay”. In: *Physical Review Letters* 108 (2012), p. 171803. DOI: [10.1103/PhysRevLett.108.171803](https://doi.org/10.1103/PhysRevLett.108.171803). arXiv: [1203.1669](https://arxiv.org/abs/1203.1669) [hep-ex].
- [4] Daya Bay Collaboration, F. P. An, et al. “Spectral Measurement of Electron Antineutrino Oscillation Amplitude and Frequency at Daya Bay”. In: *Physical Review Letters* 112 (2014), p. 061801. DOI: [10.1103/PhysRevLett.112.061801](https://doi.org/10.1103/PhysRevLett.112.061801). arXiv: [1310.6732](https://arxiv.org/abs/1310.6732) [hep-ex].
- [5] C. Giunti. “Neutrino Wave Packets in Quantum Field Theory”. In: *Journal of High Energy Physics* 11 (2002), p. 017. DOI: [10.1088/1126-6708/2002/11/017](https://doi.org/10.1088/1126-6708/2002/11/017). arXiv: [hep-ph/0205014](https://arxiv.org/abs/hep-ph/0205014).
- [6] KamLAND Collaboration, S. Abe, et al. “Precision Measurement of Neutrino Oscillation Parameters with KamLAND”. In: *Physical Review Letters* 100 (2008), p. 221803. DOI: [10.1103/PhysRevLett.100.221803](https://doi.org/10.1103/PhysRevLett.100.221803). arXiv: [0801.4589](https://arxiv.org/abs/0801.4589) [hep-ex].
- [7] KamLAND Collaboration, T. Araki, et al. “Measurement of Neutrino Oscillation with KamLAND: Evidence of Spectral Distortion”. In: *Physical Review Letters* 94 (2005), p. 081801. DOI: [10.1103/PhysRevLett.94.081801](https://doi.org/10.1103/PhysRevLett.94.081801). arXiv: [hep-ex/0406035](https://arxiv.org/abs/hep-ex/0406035).
- [8] A. Kobach, A. V. Manohar, and J. McGreevy. “Neutrino Oscillation Measurements Computed in Quantum Field Theory”. In: *Physics Letters B* 783 (2018), pp. 59–75. DOI: [10.1016/j.physletb.2018.06.021](https://doi.org/10.1016/j.physletb.2018.06.021). arXiv: [1711.07491](https://arxiv.org/abs/1711.07491) [hep-ph].
- [9] K. Kraus. “General State Changes in Quantum Theory”. In: *Annals of Physics* 64 (1971), pp. 311–335. DOI: [10.1016/0003-4916\(71\)90108-4](https://doi.org/10.1016/0003-4916(71)90108-4).
- [10] Z. Maki, M. Nakagawa, and S. Sakata. “Remarks on the Unified Model of Elementary Particles”. In: *Progress of Theoretical Physics* 28.5 (1962), pp. 870–880. DOI: [10.1143/PTP.28.870](https://doi.org/10.1143/PTP.28.870).
- [11] A. J. Maley. “Accidental Symmetry by Operator Dimension”. Manuscript, 2026. 2026.
- [12] A. J. Maley. “Flavor Symmetry, Basis Skin, Spurion Structure, and Invariant Normal Form”. Manuscript, 2026. 2026.
- [13] A. J. Maley. “Flavor Transition, Mixing, and Generation Cardinality: CKM/PMNS Coordinates, CP Phases, Basis Skins, Role-Cardinality Burdens, and No Primitive Mixing-Matrix Identity in the Standard Model”. Manuscript, 2026. 2026.
- [14] A. J. Maley. “Generation Cardinality as a Forced Role-Occupancy Endpoint”. Manuscript, 2026. 2026.
- [15] A. J. Maley. “Non-Degenerate Construction and the Kernel of Admissibility: An Exhaustion Theorem for Standing, Reference, and Irreversibility”. Manuscript, 2 August 2026. 2026.
- [16] A. J. Maley. “Role-Occupancy Closure under Admissibility: Why Role-Cardinality Matching Is Only the Cardinality Shadow of Compatible Realization”. Manuscript, May 2026. 2026.
- [17] A. J. Maley. “Simultaneous Constraint Collapse of the Admissible Gauge Arena: Arena-Rigidity Foundations for Standard-Model Realization under AASC”. Manuscript, May 2026. 2026.
- [18] A. J. Maley. “The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissibility-Bearing Construction”. Manuscript, 16 February 2026. 2026.

- [19] A. J. Maley. “The Realized Physical Interior: Locality, Metric Role, Fields, Particles, Gauge Structure, Mass, Vacuum, Time, Causality, and the Standard Physics Stack under AASC”. Manuscript, June 2026. 2026.
- [20] A. J. Maley. “The Standard Model Carrier Exhaustion Closure”. Manuscript, 2026. 2026.
- [21] A. J. Maley. “The Standard Model Symmetry Atlas and Exhaustion Closure”. Manuscript, 2026. 2026.
- [22] A. J. Maley. “Unus Solus Possibilis Est: Corpus Closure and Uniqueness of the Admissible Interior”. Manuscript, 9 February 2026. 2026.
- [23] S. P. Mikheyev and A. Y. Smirnov. “Resonant Amplification of Neutrino Oscillations in Matter and Solar-Neutrino Spectroscopy”. In: *Soviet Journal of Nuclear Physics* 42 (1985). Original Russian publication: *Yadernaya Fizika* 42, 1441–1448, pp. 913–917.
- [24] OPERA Collaboration, N. Agafonova, et al. “Discovery of Tau Neutrino Appearance in the CNGS Neutrino Beam with the OPERA Experiment”. In: *Physical Review Letters* 115 (2015), p. 121802. doi: [10.1103/PhysRevLett.115.121802](https://doi.org/10.1103/PhysRevLett.115.121802). arXiv: [1507.01417 \[hep-ex\]](https://arxiv.org/abs/1507.01417).
- [25] SNO Collaboration, Q. R. Ahmad, et al. “Direct Evidence for Neutrino Flavor Transformation from Neutral-Current Interactions in the Sudbury Neutrino Observatory”. In: *Physical Review Letters* 89 (2002), p. 011301. doi: [10.1103/PhysRevLett.89.011301](https://doi.org/10.1103/PhysRevLett.89.011301). arXiv: [nuc1-ex/0204008](https://arxiv.org/abs/nuc1-ex/0204008).
- [26] W. F. Stinespring. “Positive Functions on C*-Algebras”. In: *Proceedings of the American Mathematical Society* 6.2 (1955), pp. 211–216. doi: [10.1090/S0002-9939-1955-0069403-4](https://doi.org/10.1090/S0002-9939-1955-0069403-4).
- [27] Super-Kamiokande Collaboration, K. Abe, et al. “Evidence for the Appearance of Atmospheric Tau Neutrinos in Super-Kamiokande”. In: *Physical Review Letters* 110 (2013), p. 181802. doi: [10.1103/PhysRevLett.110.181802](https://doi.org/10.1103/PhysRevLett.110.181802). arXiv: [1206.0328 \[hep-ex\]](https://arxiv.org/abs/1206.0328).
- [28] Super-Kamiokande Collaboration, Y. Ashie, et al. “Evidence for an Oscillatory Signature in Atmospheric Neutrino Oscillation”. In: *Physical Review Letters* 93 (2004), p. 101801. doi: [10.1103/PhysRevLett.93.101801](https://doi.org/10.1103/PhysRevLett.93.101801). arXiv: [hep-ex/0404034](https://arxiv.org/abs/hep-ex/0404034).
- [29] Super-Kamiokande Collaboration, Y. Fukuda, et al. “Evidence for Oscillation of Atmospheric Neutrinos”. In: *Physical Review Letters* 81 (1998), pp. 1562–1567. doi: [10.1103/PhysRevLett.81.1562](https://doi.org/10.1103/PhysRevLett.81.1562). arXiv: [hep-ex/9807003](https://arxiv.org/abs/hep-ex/9807003).
- [30] S. Weinberg. “Baryon- and Lepton-Nonconserving Processes”. In: *Physical Review Letters* 43 (1979), pp. 1566–1570. doi: [10.1103/PhysRevLett.43.1566](https://doi.org/10.1103/PhysRevLett.43.1566).
- [31] L. Wolfenstein. “Neutrino Oscillations in Matter”. In: *Physical Review D* 17 (1978), pp. 2369–2374. doi: [10.1103/PhysRevD.17.2369](https://doi.org/10.1103/PhysRevD.17.2369).